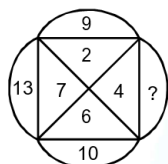




1.



(Figure Puzzle) NIMCET-2007

- (a)9 (b)6 (c)10 (d)12

2. A cube has its 6 faces having different colours(red , blue , green , yellow , white and orange). It has been given that green is opposite to red, blue is between green and red, yellow is adjacent to orange, What colours are on the adjacent faces of face having yellow colours:

- (a) green, orange, blue, red **(Puzzle) NIMCET-2007**
(b) red, blue, green, white
(c) green, blue, white, orange
(d) green, orange, red, white

Directions for Questions Q 3 to Q 6 : Read the following information carefully and answer the questions that follow.

- (i) Jayant, Kamal, Namita, Asha and Tanmay are five members of a family.
(ii) They have their birth dates from January to May, each member in one of these months.
(iii) Each one likes one particular item for his/her birthday out of Bengali Sweets, Chocolates, Pastries, Ice Cream and Dry Fruits.
(iv) The one who likes Pastries is born in the month which is exactly middle in the months given.
(v) Asha does not like Ice cream but brings Chocolates for Jayant in February.
(vi) Tanmay who is fond of Bengali sweets is born in the next month immediately after Namita. **(Puzzle) NIMCET-2007**
(vii) Namita does not like Dry fruits or Ice cream

3. What is the choice of Asha?
(a) Pastries (b) Dry fruits
(c) Bengali sweets (d) Cannot be determined
4. Which combination of month and item is true for Jayant?
(a) March — Pastries (b) February — Pastries
(c) None of these (d) Cannot be determined
5. What is the choice of Kamal?
(a) Ice cream (b) Bengali sweets
(c) Dry fruit (d) Cannot be determined
6. In which month was Kamal born ?
(a) January (b) May
(c) January or May (d) Data inadequate

Directions for Questions Q.7 to Q.11 : (Puzzle) NIMCET-2007

Read the following information carefully and answer the questions that follow :

- (i) There are eight faculty members A, B, C, D, E, F, G and H in the institute, each teaching a different subject.
(ii) There are three lady members and of the eight, four are holding Ph.D Degree.
(iii) E teacher Psychology and is Ph.D. A teachers Chemistry.
(iv) The one who teachers Economics is not Ph.D. No lady member teachers either Commerce or Law. Law faculty does not award Ph.D.
(v) D and G do not teach either Commerce or Physics.
(vi) H and C are lady members and are not Ph.D. F who is Ph.D. teachers Zoology.
(vii) B and G are Ph.Ds and G is a lady member.

7. Who teaches Physics ?

- (a)C (b) Either H or C
(c) C and D (d) Cannot be determined

8. Which of the following lady members is/are Ph.D. ?

- (a)G (b) G and H
(c) C and D (d) Cannot be determined

9. Which of the following statements is true ?

- (a) Two lady members are Ph.D
(b) Three male members are Ph.D
(c) The person who teacher Zoology is not Ph.D.
(d) The person who teacher Economics is Ph.D.

10. Which of the following combination is not correct ?

- (a) Commerce -Male - Ph.D
(b) Economics-Lady-Non-Ph.D.
(c) Physics-Lady-Ph.D.
(d) Zoology-Male-Ph.D.

11. What is the subject taught by G ?

- (a) Zoology
(b) Either Physics or Zoology
(c) Either Physics or Economics
(d) Cannot be determined

12. You are given two (unmarked) containers of capacity 9 and 4 L and a huge tank of water. Need is to get a measure of exactly 6 L of water. A move is either filling a container completely or emptying a container (either fully or partially). The smallest number of moves needed to do this take is **(Puzzle) NIMCET-2008**

- (a)8 (b)10 (c)12 (d)None

13. What is the diameter of the largest circle that can be drawn on a chessboard so that its entire circumference gets covered by the black squares and no part of the circumference on any white space, given that the chessboard has black and white squares of size one inch. **(Puzzle) NIMCET-2008**

- (a)1 inch (b) $\sqrt{2}$ inch
(c) $\sqrt{10}$ inch (d) $2\sqrt{3}$ inch

14. Four friends-Arjan, Bhuvan, Guran and Lakha comparing the number of sheep that they owned. It was found that Guran had ten more sheep than Lakha. If Arjan gave one-third to Bhuvan and Bhuvan gave a quarter of what he then held to Guran, who then passed on a fifth of his holding to Lakha, they would all have an equal number of sheep. How many sheep did each of them possess? Give the minimal possible answer. **(Puzzle) NIMCET-2008**

- (a)200, 105, 110, 100 (b)90, 55, 55, 45
(c)180, 110, 110, 100 (d)None of these

15. Three players-Aalu, Kachaalu and Bhalu were playing poocker and suddenly started to quarrel among themselves blaming each other for cheating. It was found out that atleast one person among the three cheated. When they were asked who cheated, their replies were as follows. **(Puzzle) NIMCET-2008**

Aalu I did not cheat, Kachaalu cheated

Kachaalu I did not cheat, Both Aalu and Bhalu cheated.

Bhalu I did not cheat, only Kachaalu did not cheat.

If exactly one person among them always spoke truth, another always lied and the third alternated between the truth and lie, then which of the following statements can never be true is any case?

- (a) Only Aalu and Bhalu cheated
(b) Only Aalu and Bhalu did not cheat





- (c) Bhalu always spoke the truth
(d) Bhalu alternated between truth and lie.

Direction (Q. Nos. 16-19) Read the following statements and answer questions.

The office staff of XYZ corporation presently consists of three book keepers, P, Q, R and 5 secretaries S, T, U, V, W. the management is planning to open a new office in another city using 2 book keepers and 3 secretaries of the present staff. To do so they plan to separate certain individuals who don't function well together. The following guidelines were established to set up the new office. **(Puzzle) NIMCET-2008**

- (i) Book keepers P and R are constantly finding fault with one another and should not be sent together to the new office as a team.
 - (ii) R and T function well alone but as a team, they should be separated.
 - (iii) S and V have not been on speaking terms and shouldn't go together.
 - (iv) Since, S and U have been competing for promotion they shouldn't be a team
16. If P is to be moved as one of the bookkeepers, which of the the following cannot be a possible working unit?
(a) PQSTW (b) PQSVW
(c) PQTUW (d) PQTUV
17. If R and U are moved to the new office, how many combinations are possible?
(a)1 (b)2 (c)3 (d)4
18. If R is sent to the new office, which member of the staff cannot go with R?
(a)Q (b)S (c)W (d)V
19. If S goes to the new office, which of the following is true?
(a)Only R cannot go (b)Only P cannot go
(c)Only P and R cannot go (d)R cannot go and W must go
20. Substitute digits for the letters to make the following relation true.

$$\begin{array}{r} \text{S T I L L} \\ + \text{W I T H I N} \\ \hline \text{L I M I T S} \end{array}$$

(Puzzle) NIMCET-2008

Note: The leftmost letter can't be zero in any word. Also, there must be a one-to-one mapping between digits and letters. e.g., if you substitute 3 for the letters S, no other letter can be 3 and all other S in the puzzle must be 3.

- (a) $98533 + 258056 = 356589$
(b) $41211 + 527013 = 938224$
(c) $98533 + 158056 = 256589$
(d) $47166 + 517013 = 614179$
21. In the middle of the confounded desert, there is the lost city of "Ash". To reach it, I will have to travel overland by foot from the coast. On a trek like this, each person can only carry enough rations for five days and the farthest we can travel in one day is 30 miles. Also, the city is 120 miles from the starting point. What I am trying to figure out is the fewest number of persons, including myself, that I will need in our group so that I can reach the city, stay overnight, and then return to the coast, without running out of supplies. How many persons (including myself) will I need to accomplish this mission?
(Puzzle) NIMCET-2008
(a)5 (b)6 (c)4 (d)3
22. A woman took a certain number of eggs to the market and sold some of them. The next day through her poultry industry,

the number left over had been doubled, and she sold the same number as the previous day. On the third day, the new remainder was tripled, and she sold the same number as before. On the fourth day, the remainder was quadrupled, and her sale were the same as before. On the fifth day, what had been left over were quintupled, yet she sold exactly the same as on all the previous occasions and so disposed of her entire stock. What is the smallest number of eggs she could have taken to the market the first day, and how many did she sell daily?
(Puzzle) NIMCET-2008

- (a)110.50 (b)127.65
(c)100.60 (d)103.60

23. The Bulls, Pacers, Lakers and Jazz ran for a contest. Anup, Sujit, John made the following statements regarding results.
- Anup said either Bulls or Jazz will definitely win.
 - Sujit said he is confident that Bulls will not win.
 - John said he is confident that neither Jazz nor Lakers will win.

When the results came, it was found that only one of the above three had made a correct statement. Who has made the correct statement and who has won the contest?

(Puzzle) NIMCET-2008

- (a) Anup, Bulls (b) John, Pacers
(c) Sujit, Lakers (d) Sujit, Jazz

24. Examine the following sequence of numbers

1
1 1
2 1
1 2 1 1
1 1 1 2 2 1
3 1 2 2 1 1
1 3 1 1 2 2 2 1
1 1 1 3 2 1 3 2 1 1

3 1 1 3 1 2 1 1 1 3 1 2 2 1 **(Puzzle) NIMCET-2008**

What are the next two numbers in the given series?

- (a)13211311122111231131 and 11213211321222111131221133
(b)23113112211132113111 and 1121321132122211131221133
(c)11231131122111321131 and 1131221212221133112132113
(d)13211311123113112211 and 113122113311213211321221

25. There were two men standing on a street. The one says to the other. "I have 3 daughters, the product of their ages is 36. What is the age of of the OLDEST daughter?"
The second guy says, "I need more information. "So, the first guy says, "The sum of their ages is equal to the address of the house across the street."
The second guy looks at the address and says, "I still need more information". So, the first guy says. "My oldest daughter wears a red dress." **(Puzzle) NIMCET-2008**
(a)9 (b)6 (c)12 (d)4
26. Three Gold (G) coins, three Silver (S) coins and three Copper (C) coins are arranged in a single row as follows
- Only 2 adjacent unlike coins can be moved at any one time.
 - The moved coins must be in contact with atleast one other coin in line i.e., no pair of coins is to be moved and placed away from the remaining ones.





- No coin pairs can be reversed i.e., S-C combination must remain in that order in its new positions when it is moved.

What is the minimum number of moves required to get all the coins in following order? **(Puzzle) NIMCET-2008**

C C C S S S G G G

- (a)6 (b)9 (c)8 (d)12

27. Mr. and Mrs. Birla and Mr. and Mrs. Tata competed in a chess tournament. Of the three games played

- (I) In only the first game were the two players married to each other.

- (II) The men won two games and the women won one game.

- (III) The Birlas won more games than the Tatas.

- (IV) Anyone who lost a game did not play a subsequent game.

Who did not lose a game? **(Puzzle) NIMCET-2008**

- (a)Mr Birla (b)Mrs Birla

- (c)Mr Tata (d)Mrs Tata

28. Larger, medium and small ships are used to bring water. 4 large ships carry as much water as 7 small ships. 3 medium ships carry the same amount of water as 2 larger ships and 1 small ship. 15 large, 7 medium and 14 small ships, each made 36 journeys and brought a certain quantity of water. In how many journeys would 12 large, 14 medium and 21 small ships bring the same quantity of water? **(Puzzle) NIMCET-2008**

- (a)32 (b)25 (c)29 (d)49

29. Five men P, Q, R, S and T read newspaper. The one who reads first gives it to R. the one who reads last had taken it from P. T was not the first or the last to read. There were two readers between Q and P. To whom did Q pass the newspaper?

(Puzzle) NIMCET-2008

- (a)R (b)P (c)S (d)T

Directions (Q. Nos. 30-34) Read the following passage and answer the questions.

Sports (and game) persons P, Q, R, S, T, U and V of a university are at the Bengaluru Airport. Five of them are selected players and leaving to participate in the grand sports Event in five different events cricket, carrom, badminton and table tennis being held at 5 different cities Mumbai, Chennai, Kolkata, Delhi and Hyderabad.

- (i) P is going to Delhi, but he does not play either cricket or carrom
(ii) Q has come to give send off to R, who is a chess player and is not leaving to either Mumbai or Hyderabad.
(iii) S is leaving to Kolkata to play table tennis.
(iv) T is not a selected player. **(Puzzle) NIMCET-2008**

30. Who plays badminton?

- (a)P (b)Q (c)R (d)S

31. Cricketer goes to

- (a)Mumbai (b)Hyderabad
(c)Chennai (d)Delhi

32. Player of which game goes to Delhi?

- (a)Badminton (b)Chess
(c)Cricket (d)Table tennis

33. Who plays chess and where is he going?

- (a)R and Chennai (b)S and Mumbai
(c)U and Delhi (d)None of these

34. There are six houses in a row. Mr. Lal has Mr. Babu and Mr. Anil as neighbours. Mr. Bhatia has Mr. Gupta and Mr. Sharma as neighbours. Mr. Gupta's houses is no next to Mr. Babu or Mr. Anil and Mr. Sharma does not live next to Mr. Anil. Who are Mr. Babu's next-door neighbours? **(Puzzle) NIMCET-2008**

- (a)Mr. Lal and Mr. Bhatia (b)Mr. Lal and Mr. Anil
(c)Mr. Sharma and Mr. Lal (d)Only Mr. Lal

35. Bala had three sons. He had some chocolates which he distributed among them. To his eldest son, he gave 3 chocolates more than half the number of chocolates with him. To his second eldest son, he gave 4 chocolates more than one-third of the remaining number of chocolates more with him. To his youngest son, he gave 4 chocolates more than one-fourth of the remaining number of chocolates with him. He was left with 11 chocolates. How many chocolates did he initially have?

(Puzzle) NIMCET-2009

- (a)180 (b)78 (c)144 (d)120

36. Twelve villages in a district are divided into 3 zones with 4 villages per each zone. The telephone department of the district intends to connect the villages with telephone lines such that every two villages in the same zone are connected with three direct lines and every two villages belonging to different zones are connected with two direct lines. How many direct lines are required? **(Puzzle) NIMCET-2009**

- (a)210 (b)96 (c)54 (d)150

37. Cars are safer than planes. Fifty per cent of plane accidents results in death, while only one percent of car accidents result in death.

Which of the following, if true, would most seriously weaken the argument above? **(Puzzle) NIMCET-2009**

- (a) Planes are inspected more often than cars.
(b) The number of car accidents is several hundred thousand times higher than the number of plane accidents.
(c) Pilots never fly under the influence of alcohol, while car drivers often do.
(d) Plane accidents are usually the fault of air traffic controllers, not of pilots.

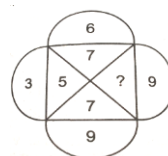
38. Reena visited her High School friend, Natasha after their 25th school reunion. "what a nice pair of children you have, are they twins?" Reena asked.

"No my sister is older than I", said Natasha's son Rahul. "The square of my age plus the cube of her age is 7148". "The square of my age plus the cube of his age is 5274", said Preeti, Natasha's daughter.

How old were they? **(Puzzle) NIMCET-2009**

- (a)Preeti 23, Rahul 14 (b)Preeti 18, Rahul 16
(c)Preeti 21, Rahul 19 (d)Preeti 19, Rahul 17

39. Which number will be there in the place of question mark (?) in the following figure? **(Puzzle) NIMCET-2009**



- (a)5 (b)6 (c)4 (d)8

40. You are in the land logic, where there are 3 types of rabbits. Blue rabbits always tell the truth, green rabbits sometimes tell the truth and red rabbits never tell the truth. Assume you cannot distinguish colours. A rabbit says to you "I always lie". What colour of rabbit is speaking to you? **(Puzzle) NIMCET-2009**

- (a) Blue (b)Red
(c)Green (d)Cannot be concluded





41. It is wrong for doctors to lie about their patients illnesses? Aren't doctors just like any other people we hire to do a job for us? Surely, we would not tolerate not being told the truth about the condition of our automobile from the mechanic we hired to fix it, or the condition of our roof from the carpenter we employed to repair it. Just as these workers would be guilty of violating their good faith contracts with us if they were to do this, doctors who lie to their patients about their illnesses violate these contracts as well, and this is clearly wrong.

The conclusion of the argument is best expressed which of the following? **(Puzzle) NIMCET-2009**

- (a) Doctor who lie to their patients about their illnesses violate their good faith contracts with their patients.
- (b) Doctor often lie to their patients about their patients illnesses.
- (c) It is wrong for doctors to lie about their patients illnesses.
- (d) Doctor, like mechanics and carpenters, enter into good faith contracts with us when we hire them.

42. Recently, while in Bangalore, I decided to walk down the escalator of a tube station. I did some quick calculation in my mind. I found that if I walk down twenty six steps, I require thirty seconds to reach the bottom. However, if I am able to step down thirty four stairs I would only require eighteen seconds to get to the bottom. If the time is measured from the moment the top step begins to descend to the time I step off the last step at the bottom, what is the height of the stairway in steps? **(Puzzle) NIMCET-2009**

- (a)40 (b)46 (c)52 (d)58

Directions (Q. Nos. 43-46) Read the following information carefully to answer the questions.

After months of talent searching for an administrative assistant to the president of the college the field of applicants had been narrowed down to five (A, B, C, D, and E). It was announced that the finalist would be chosen after a series of all-day group personal interviews. The examining committee agreed upon the following procedure.

- (1) The interviews will be held once a week.
 - (2) Three candidates will appear at an all-day interview session.
 - (3) Each candidate will appear atleast once.
 - (4) If it becomes necessary to call applicants for additional interviews.
 - (5) Because of a detail given in the written applications, it was agreed that whenever candidate B appears, A should also be present.
 - (6) Because of travel difficulties, it was agreed that C will appear for only one interview.
43. Which of the following is a possible sequence of combinations for interviews in two successive weeks?
- (a) ABC ; BDE (b) ABD ; ABE
 - (c) ADE ; ABC (d) BDE ; ACD
44. At the first interview, the following candidates appear: A, B and D. Which of the following combinations can be called for the interview to be held the next week?
- (a)BCD (b)CDE
 - (c)ABE (d)ABC
45. Which of the following correctly state (s) the procedure followed by the search committee?
- (I) After the second interview, all applicants might have appeared atleast once.

- (II) The committee interviews each applicant asecond time.
- (III) If a third session is held, it is possible for all applicants to appear atleast twice.

- (a) I only (b) II only
- (c) I and II only (d) III only

46. A, B and D appear at the interview and D is called for an additional interview the following week, which two candidates may be asked to appear with D?

- (i)A (II)B (III)C (IV)E
- (a)I and II only (b)III and IV only
- (c)II and III only (d)II and IV only

Direction (Q. Nos. 47-50) Read the following passage to answer the questions

Rajita has a unique way of attempting the question paper having 50 questions. She starts from question 1 and attempts all questions which are in A.P. with a commorl difference of 3 in the forward direction and 3 in the reverse direction. If she reaches a stage when she cannot attempt any more question, she starts in the reverse direction with the first unanswered question. She repeats the same process and when she reaches a stage when she can not process any further, she reverses her direction again starting with the first unanswered question. **(Puzzle) NIMCET-2009**

47. How many times does she reverse her direction?

- (a)3 (b)4 (c)5 (d)6

48. Which is the last question that she answers if she at- tempts all the 50 questions?

- (a)50 (b)49 (c)48 (d)3

49. Which is the 20th question Rajita answers ?

- (a)50 (b)48 (c)47 (d)44

50. If $A_1 = \{3\}$, $A_2 = \{5, 7, 9\}$, $A_3 = \{11, 13, 15, 17, 19\}$, $A_4 = \{21, 23, 25, 27, 29, 31, 33\}$ and so on, what is the average of the numbers of the set A_{20} ? (Average)

- (a)761 (b)763 (c)765 (d)767

Direction (Q. Nos. 51-55) Read the following passage to answer the questions

Five houses lettered A, B, C, D, & E are built in a row next to each other. The houses are lined up in the order A, B, C, D, & E. Each of the five houses have coloured roofs and chimneys. The roof and chimney of each house must be painted as follows

- (1) The roof must be painted either green, red, or yellow.
- (2) The chimney must be painted either white, black~or red.
- (3) No house may have the same colour chimney as the colour of roof.
- (4) No house may use any of the same colours that the every next house uses.
- (5) House E has a green roof.
- (6) House B has a red roof and a black chimney.

(Puzzle) NIMCET-2009

51. Which statement is false ?

- (a)House A has a yellow roof.
- (b)House A & C have different colour chimneys.
- (c)House D has a black chimney.
- (d)House E has a white chimney.

52. Which of the following is true ?

- (a) At least two houses have black chimneys.
- (b) At least two houses have red roofs.
- (c) At least two houses have white chimneys.
- (d) At least two houses have green roofs.

53. What is maximum total number of green roofs for houses





(a)1 (b)2 (c)3 (d)4

54. Which possible combinations of roof & chimney can a house have ?

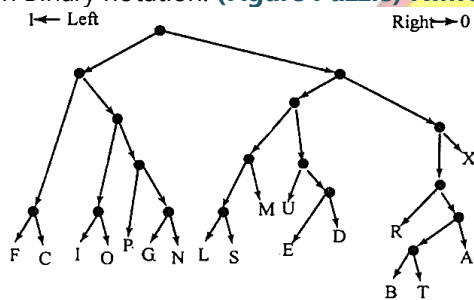
- (I) A red roof & a black chimney.
(II) A yellow roof & a red chimney.
(III) A yellow roof & a black chimney.

(a) I & II & III (b) II only (c) III only (d) I & II only

55. If house C has a yellow roof, which one of the following is true?

- (a) House E has a white chimney.
(b) House E has a black chimney.
(c) House E has a red chimney.
(d) House D has a red chimney.

Direction : Questions 56 to 59 : Given below is a binary tree, where every letter has been coded with a string of digit 0 and 1. At any node going left is denoted by 1; at any node going right is denoted by 0. Thus N is denoted as : 10000. All the codes are in Binary notation. **(Figure Puzzle) NIMCET-2010**



56. What will be the code for S :

- (a) 01011 (b) 01110
(c) 01111 (d) None

57. Which letter is represented by 11001 ?

- (a) G (b) L (c) U (d) None

58. What is the value of C + R in binary notation ?

- (a) 11101 (b) 1101 (c) 1001 (d) None

59. If all the codes are converted into decimal notation, then how many letters have their value greater than L ?

- (a) 1 (b) 2 (c) 3 (d) None

Directions : Questions 60 to 63 : 220 guests are to be transported from A to B. Any number of buses of the following passenger carrying capacities are available.

(PUZZLE) NIMCET-2010

Type P : 60, **Type Q :** 50, **Type R :** 40, **Type S :** 30

The cost per trip for a bus of each of these types is given as follows :

Type P : Rs 200, **Type Q :** Rs 140, **Type R :** Rs 125, **Type S :** Rs 95

No buses can be overloaded and prefer no vacant seats in each trips.

60. What is the minimum possible cost for the trip ?

- (a) Rs 690 (b) Rs 615
(c) Rs 640 (d) Rs 695

61. How many buses are needed for the above (Minimum cost trip)

- (a) 5 (b) 4 (c) 7 (d) 6

62. The second cheapest trip arrangement would involve

- (a) Rs 630 (b) Rs 680 (c) Rs 710 (d) Rs 655

63. In the following square, numbers have been filled according to some rule. One space has been left blank. Find the correct number out of those given below for the blank space. **(Number Puzzle) NIMCET-2010**

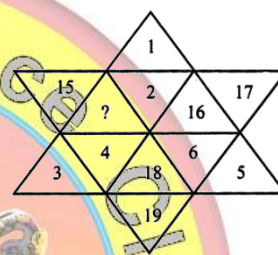
56	65	78
12	?	30
44	14	48

- (a) 14 (b) 44
(c) 62 (d) 51

Directions : Questions 64 and 65 : Find the missing number in each of the following questions :

(Number Puzzle) NIMCET-2010

64.



- (a) 13
(c) 20

- (b) 14
(d) 21

65.



- (a) 10
(c) 12

- (b) 11
(d) 13

(Number Puzzle) NIMCET-2010

Direction : Question 66 to 68 : Read the following information carefully and answer the question that follow :

- (i) There is group of five person – P, Q, R, S and T
(ii) One of them is horticulturist, one is a physicist, one is a journalist, one is an industrialist and one is an advocate.
(iii) Three of them – P, R and advocate prefer tea to coffee and two of them – Q and the journalist prefer coffee to tea.
(iv) The industrialist, S and P are friends to one another but two of these prefer coffee to tea
(v) The horticulturist is R's brother.

(Puzzle) (Number Puzzle) NIMCET-2010

66. Who is a horticulturist ?

- (a) P (b) Q (c) R (d) S

67. Who is an industrialist ?

- (a) T (b) R (c) Q (d) S

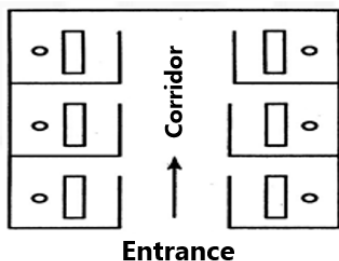
68. Which of the following groups include a person who likes tea but is not an advocate ?

- (a) PRT (b) ST (c) QRT (d) None



Directions: Questions 69 to 72 : The plan given below, shows office for six officers namely A, B, C, D, E, and F. Both B and C occupy offices to the right of the corridor (as one enters the office block) and A occupies the office to the left of the corridor. E and F occupy offices on opposite sides of the corridor but their offices do not face each other. The offices of C and D face each other. E does not have a corner office. F's office is further down the corridor than A's, but on the same side.

(Puzzle) NIMCET-2010



69. If E sits in his office and faces the corridor, whose office is to his left ?
(a)A (b)B (c)C (d)D
70. Whose office faces A's office ?
(a)B (b)C (c)D (d)E
71. Who is/are F's neighbour(s) ?
(a)A only (b)A and D (c)C only (d)B and C
72. D was heard telling someone to go further down the corridor to the last office on the right. To whose room was he trying to direct that person ?
(a)A (b)B (c)C (d)F
73. Persons X, Y, Z and Q live in red, green, yellow or blue colored houses placed in a sequence on a street. Z lives in a yellow house. The green house is adjacent to the blue house. X does not live adjacent to Z. The yellow house is in between the green and red house. The color of the house X lives in is **(Puzzle) NIMCET-2010**
(a)Green (b)Blue (c)Red (d)none

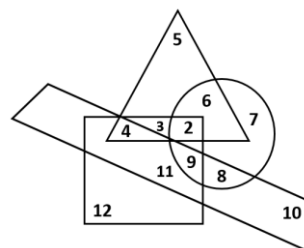
Directions : Q. 74-76 : Read the information given below and answer the questions that follow :

Four persons A, B, C and D play a cards game. They put Rs. 500 as stake money. When the game is over 'C' receives Rs. 19 more than 'D' and 'B' receives Rs. 21 less than 'A' whose amount was Rs. 2 less than the quarter of Rs. 500

74. How much money did 'C' get ? **(Puzzle) NIMCET-2011**
(a)Rs. 147 (b)Rs. 136
(c)Rs. 144 (d)Rs. 159
75. How much money did 'B' get ?
(a)Rs. 102 (b)Rs. 107
(c)Rs. 108 (d)Rs. 110
76. Who get highest amount ?
(a)A (b)B (c)C (d)D

Directions : Q. 77- 79 : In the following diagram circle stands for 'educated' square for 'hardworking', triangle for 'urban people' and rectangle for 'honest'. Different regions in the

diagram are numbered from 2 to 12. Study the diagram carefully and answer. **(Puzzle) NIMCET-2011**



77. Educated, hard-working and urban people are indicated by
(a)7 (b)2 (c)3 (d)4
78. Non-urban educated people who are neither hardworking nor honest are indicated by
(a)5 (b)7 (c)10 (d)12
79. Honest, educated and hardworking non urban people are indicated by
(a)3 (b)4 (c)6 (d)9

Directions : Q. 80-83 : Six scientists A, B, C, D, E and F are to present at paper each at a one-day conference. Three of them will present their papers in the morning session before the lunch break whereas the other three will be presented in the afternoon session. The lectures have to be scheduled in such a way that they comply with the following restrictions:

B should present his paper immediately before C's presentation; their presentations cannot be separated by the lunch break. D must be either the first or the last scientist to present his paper. **(Puzzle) NIMCET-2011**

80. In case C is to be the fifth scientist to present his paper, then B must be
(a)first (b)second
(c)third (d)fourth
81. B could be placed for any of the following places in the order of presenters EXCEPT
(a)second (b)third
(c)fourth (d)fifth
82. In case F is to present his paper immediately after D presents his paper, C's could be scheduled for which of the following places in the order of presenters?
(a)second (b)third
(c)fourth (d)fifth
83. In case F and E are the fifth and sixth presenters respectively then which of the following must be true?
(a)A is first in the order of presenters
(b)A is third in the order of presenters
(c)A is fourth in the order of presenters
(d)B is first in the order of presenters

Directions : Q. 84 - 86 : (Puzzle) NIMCET-2011

Mrs. Thomes received a large order for stitching school uniforms from Mayflower school and Littleflower school. She has two cutters who will cut the fabric, five tailors who will do the stitching and two assistants to stitch the button and button holes. Each of these nine persons will work for exactly 10 hours a day. Each of the Mayflower uniforms requires 20



min. for cutting the fabric, one hour for stitching, and 15 min. for stitching buttons and button holes, whereas the Littleflower uniform requires 30 min., 1 hour and 30 min. respectively for these activities.

84. What is the number of Littleflower uniforms that Mrs. Thomes can complete in a day ?

- (a)50 (b)20 (c)40 (d)30

85. On a particular day, Mrs. Thomes decided to complete 20 Littleflower uniforms. How many Mayflower uniforms can she complete on that day ?

- (a)30 (b)40 (c)20 (d)0

86. If she hires one more assistant, what is the maximum number of Mayflower uniforms that she can complete in a day ?

- (a)40 (b)50 (c)60 (d)30

Directions (Q. Nos. 87-91) Read the following passage carefully and answer the questions. **(Puzzle) NIMCET-2012**

- A causes B or C but not both.
- F occurs only if B occurs.
- D occurs, if B or C occurs.
- E occurs only if C occurs.
- J occurs only if E or F occurs.
- D causes G or H or both.
- H occurs, if E occurs.
- G occurs, if F occurs.

87. If A occurs, which may occurs?

- I. F and G
II. E and H
III. D

- (a) Only I
(b) Only II
(c) I and III or II and III, but not both
(d) I, II and III

88. If B occurs, which must occur?

- (a) D
(b) G
(c) H
(d) J

89. If J occurs, which must have occurred?

- (a) Both E and F
(b) Either B or C
(c) Both B and C
(d) None of these

90. If $+is *$, $-is +$, $*is /$ and $/is -$, then $6-9+8*3/20$ is equal to **(Puzzle) NIMCET-2012**

- (a) -2
(b) 6
(c) 10
(d) 12

91. A study of native born residents in an area of Adivasis found that two-third of the children developed considerable levels of the nearsightedness after starting school, while their illiterate parents and grandparents,

who had no opportunity for formal schooling, showed no signs of this disability.

If the above statements are true, which of the following conclusions is most strongly supported by them? **(Puzzle) NIMCET-2012**

- (a) Only people who have the opportunity for formal schooling develop nearsightedness
(b) People who are illiterate do not suffer from nearsightedness
(c) The nearsightedness in the children is caused by the visual stress required by reading and other class work
(d) Only literate people are nearsighted

Directions (Q.Nos.92-94) Read the following passage carefully and answer the questions. **(Puzzle) NIMCET-2012**

Five roommates Randy, Sally, Terry, Uma and Vernon each do one housekeeping taskmopping, sweeping, laundry, vacuuming or dusting one day a week, Monday through Friday.

- Vernon does not vacuum and does not do his task on Tuesday.
- Sally does the dusting and does not do it on Monday of Friday.
- The mopping is done on Thursday.
- Terry does his task, which is not vacuuming, on Wednesday.
- The laundry is done on Friday and not by Uma.
- Randy does his task on Monday.

92. The task done by Terry on Wednesday is

- (a) Vacuuming
(b) Dusting
(c) Mopping
(d) Sweeping

93. The day on which the vacuuming is done, is

- (a) Friday
(b) Monday
(c) Tuesday
(d) Wednesday

94. Sally does dusting on

- (a) Friday
(b) Monday
(c) Tuesday
(d) Wednesday

Direction(Q.Nos. 95-96) Read the following passage carefully and answer the questions.

P,Q, R, S, T, U,V and W are sitting round the circle and are facing the centre. P is second to the right of T, T is the neighbour of R and V. S is not the neighbour of P,V is the neighbour of U,Q is not between S and W and W is not between U and S. **(Puzzle) NIMCET-2012**

95. Which two of the following are not neighbours?

- (a) RV
(b) UV
(c) RP



- (d) QW
96. What is the position of S?
- Between U and V
 - Second to the right of P
 - To the immediate right of W
 - Data inadequate

Directions(Q. Nos. 97-99) Read the following passage carefully and answer the questions. **(Puzzle) NIMCET-2012**

Five houses lettered A, B, C, D and E are built in a row next to each other. The houses are lined up in the order A, B, C, D and E. Each of the five houses have coloured roofs and chimneys. The roof and chimney of each houses must be painted as follows.

- The roof must be painted either green, red or yellow.
 - The chimney must be painted either white, black or red.
 - No house may have the same colour chimney as the colour of roof.
 - No house may use any of the same colours that adjacent house uses.
 - House E has a green roof.
 - House B has a red roof and a black chimney.
97. Which of the following is true?
- Atleast two houses have a black chimney.
 - Atleast two houses have red roofs.
 - Atleast two houses have white chimneys
 - Atleast two houses have green roofs
98. If house C has a yellow roof, then which of the following must be true?
- House E has a white chimney
 - House E has a black chimney
 - House E has a red chimney
 - House D has a red chimney
99. What is the maximum number of green roofs?
- 1
 - 2
 - 3
 - 4

Directions[Q. Nos. 100-102]An employee has been assigned the task of allotting offices to six of the staff members. The offices are numbered 1-6. The offices are arranged in a row and they are separated from each other by six foot high dividers. Hence, voices sounds and cigarette smoke flow easily from one office to another. Miss. Robert needs to use the telephone quite often throughout the day. Mr. Mike and Mr. Brown need adjacent offices as they need to consult each other often while working. Miss. Hardy is a senior employee and has to be allotted the office number 5, having the biggest window. Mr. Donald requires silence in the offices next to his and Mr. Tim prefers to be as away as possible from Miss. Robert. Mr. Donald are all smokers. Miss. Hardy finds tobacco smoke allergic and consecutively the offices next to hers to be occupied by non-smokers. Unless specifically stated all the employees maintain an atmosphere of silence during office hours. **(Puzzle) NIMCET 2013**

100. The ideal candidate to occupy the office farthest from Mr. Brown would be
- Miss. Hardy
 - Mr. Mike
 - Mr. Tim
 - Mr. Donald

101. The three employees who are smokers, should be seated in the offices
- 1, 2 and 4
 - 2, 3 and 6
 - 1, 2 and 6
 - 1, 2 and 3

102. The ideal office for Mr. Mike would be
- 2
 - 6
 - 1
 - 3

Directions [Q. Nos. 103-104]: A blacksmith has five iron articles A,B,C,D and E, each having a different weight. **(Puzzle) NIMCET 2013**

- Weights twice as much as B.
- Weights four and half times as much as C.
- Weights half as much as E.
- Weights less than A but more than C.

103. Which of the following article is heaviest in weight
- A
 - B
 - C
 - D

104. Which of the following represents the descending order of weights of the articles?

- A,B,E,D,C
- A,B,C,D,E
- B,D,E,A,C
- C,D,E,B,A

Directions [Q. nos. 105-107]: There are three switches A,B and C which can be in ON/OFF position. Their settings change as per the following rules: **(Puzzle) NIMCET 2013**

- If A is the only switch as ON, change B to ON.
 - If A and B are only switches as ON, change C to ON.
 - If all three switches are ON, change C to OFF.
 - For all other situations, all switches in On are changed to OFF and all switches in OFF are changed to ON.
105. If switches A and B are ON and C is OFF, then their changed settings will be
- A ON, B OFF, C OFF
 - A ON, B ON, C ON
 - A ON, B OFF, C ON
 - A OFF, B ON, C OFF
106. If only B is ON, then the changed setting, then which of the following could have been the original setting?
- A ON, B ON, C ON
 - A ON, B ON, C OFF
 - A ON, B OFF, C ON
 - A OFF, B OFF, C ON
107. If only B is ON in the changed setting, then which of the following could have been the original setting?
- A ON, B OFF, C OFF
 - A ON, B ON, C ON
 - A ON, B OFF, C ON
 - A OFF, B OFF, C OFF

108. Four players A, B, C and D have to form into two pairs, however, no pair can play together more than seven times in a row A and B have played seven games in a row. C and D have three in a row. C does not to work with A. Who should play with B?

(Puzzle) NIMCET-2014

- A
- D
- C
- Cannot be determined

Direction[Q. Nos. 109 to 112] Cricket club in five towns A, B, C, D and E have one team each named P, Q, R, S and T, not necessarily in the same order. **(Puzzle) NIMCET-2014**
The team in A has beaten R, P and S, Q has beaten the teams in E, C and A. Team R is in B and the team in C is not S.

109. Where is the team Q?
- A
 - B
 - C
 - D
110. Where is the team P?
- A
 - B
 - C
 - D





111. Which team is in A?
(a)P (b)Q (c)S (d)T
112. There are five different boxes of different unknown weights each less than 100 kg. These boxes were weighted in pairs and the weights obtained are 110, 112, 113, 114, 115, 116, 117, 118, 120 and 121 kg. What is the weighted in kg of the heaviest box ? **(Puzzle) NIMCET-2014**
(a)60 (b)62 (c)64 (d)61

Direction[Q. Nos. 113 to 117] : Six members of a family A, B, C, D, E and F are Psychologist, Manager, Advocate, Jeweller, Doctor and Engineer but not necessarily in same order. Doctor is the grandfather of F, who is Psychologist. Manager D is married to A. Jeweller C is married to Advocate. B is the mother of F and E. There are two married couples in the family.

(Puzzle) NIMCET-2014

113. Which is the profession of A?
(a)Manager (b)Engineer
(c)Can't be determined (d)None of these
114. What is the profession of E?
(a)Manager (b)Engineer
(c)Doctor (d)None of these
115. How is A related to E?
(a)Grandmother (b)Wife
(c)Grandfather (d)None of these
116. How many male members are there in the family?
(a)Two (b)Three
(c)Four (d)Can't be determined
117. Who are the two couples in the family?
(a)AD and CB (b)AB and CD
(c)AC and BD (d)None of these

Direction[Q. Nos. 118 to 120] : At a small company, parking spaces are reserved for the top executives: CEO, President Vice-President, Secretary and Treasurer with the spaces lined up in that order. The parking lot guard can tell at a glance, if the cars are parked correctly by looking at the colour of the cars. The cars are yellow, green, purple, red and blue and the executive names are Alice, Bert, Cheryl, David and Enid.

The car in the first space is red.

A blue car is parked between the red car and the green car.

The car in the last space is purple.

The secretary drives a yellow car.

Alice's is parked next to David's.

Enid drives a green car.

Bert's car is parked between Cheryl's and Enid's.

David's car is parked in the last space.

(Puzzle) NIMCET-2014

118. Who is the secretary ?
(a)Enid (b)David
(c)Cheryl (d)Alice

119. Who is the CEO?
(a)Alice (b)Bert
(c)Cheryl (d)David
120. What colour is the Vice-President car?
(a)Green (b)Yellow
(c)Blue (d)Purple

Directions [Q. Nos. 121-124] A circular field with inner radius of 10 m and outer radius of 20 m is divided into 5 successive stage for ploughing. The ploughing at each stage, with starting points P1, P2, P3, P4 and P5 was is allotted to one of the five farmers F1, F2, F3, F4 and F5 not necessarily in that order.

(Puzzle) NIMCET-2015

- F5 was allotted the stage starting a point P4.
 - The stage from P5 to P3 was not the first stage.
 - F4 was allotted the work of the fourth stage.
 - Finishing point of stage 3 was P1 and the work was not allotted to F1.
 - F3 was allotted the work of stage ending at P5.
121. Which of the following is the finish point for farmer F2?
(a)P1 (b)P2
(c)P3 (d)P4
122. Which stage was ploughed by F5
(a)2 (b)3
(c)4 (d)5
123. What are the starting and ending points of the field ploughed by F4?
(a)P1 and P2 (b)P1 and P4
(c)P4 and P2 (d)P2 and P4
124. What is the starting point for stage 3?
(a)P2 (b)P3
(c)P4 (d)Cannot be determined

Directions [Q. Nos. 125-129] (Puzzle) NIMCET-2015

- There is a family of six members A, B, C, D, E and F.
- There are two married couples in the family and the family members represent three generations.
- Each member has a distinct choice of a colour amongst Green, Yellow, Black, Red, White and Pink.
- No lady members likes either Green or White.
- C, who likes Black colour, is the daughter-in-law of E.
- B is the brother of F and son of D and likes Pink.
- A is the grandmother of F and F does not like Red.
- Wife of the husband having a choice for Green colour likes Yellow.

125. Which of the following is the colour preference of A?
(a)Red (b)Yellow
(c)Either Red or Yellow (d)Cannot be determined
126. Which of the following could be the colour combination of one of the couples?
(a)Yellow-Red (b)Green-Black
(c)Red-Yellow (d)Yellow-Green
127. Which of the following is one of the married couples?
(a)CD (b)AC





(c)AD (d)Cannot be determined

128. Which of the following is true about F?

- (a)Brother of B (b)Sister of B
(c)Daughter of C (d)Cannot be determined

129. Three ladies X, Y and Z marry three man A, B and C, X is married to A, Y is not married to an engineer. Z is not married to a doctor, C is not a doctor and A is a lawyer. Then, which of the following statements is correct?

- (a) X is married to a doctor **(Puzzle) NIMCET-2015**
(b) Y is married to C, who is a doctor
(c) Z is married to B, who is an engineer
(d) None of the above

Directions [Q. Nos. 130-133] A, B, C, D, E, F and G are seven girls having different amount of money from among Rs.10,20,40,60,80,120 and 200 with them. They had 3 chocolates, 2 toffees and 2 lollipops together, each other having one of these seven items.

B and F do not have chocolates and they have Rs.200 and Rs.80, respectively. **(Puzzle) NIMCET-2015**

- C has Rs.60 with her and G has an amount which neither Rs.40 nor Rs.120.
 - A has Rs10 and does not have a toffee.
 - The girl having Rs.40 with her is the only one other than A to have the same type of item.
 - E and girl having Rs.20 with her have the same kind of item.
130. How much amount does G have with her?
(a) Rs.20 (b) Rs.10
(c) Rs.60 (d) None of these
131. Which of the following girl have chocolates with items?
(a) F, C, G (b) C, G, E
(c) C, G, D (d) G, D, E
132. Which of the following combination is definitely correct?
(a)C-chocolate Rs.60 (b)G- toffee-Rs.20
(c) C-chocolate Rs.40 (d)None of these
133. Which girls has Rs.40 with her?
(a)E (b)A
(c)D (d)None of these

Directions (Q. Nos. 134 & 135)Are based on the following A boy is asked to put in a basket one mango when ordered 'One', one orange when ordered 'Two'. One apple when ordered 'Three' and is asked to take out from the basket one mango and an orange when ordered 'Four'. A sequence of order is given as

12332142314223314113234 **(Puzzle) NIMCET-2016**

134. How many total fruits will be in the basket at the end of the above order sequence?

- (a) 9 (b) 8 (c) 11 (d) 10

135. How many total oranges were in the basket at the end of the above the sequence?

- (a) 1 (b) 4 (c) 3 (d) 2

Directions (Q. Nos. 136 to 139): Are based on the following A team must be selected from the ten probable players A, B, C, D, E, F, G, H, I and J. Of these, A, C, E and J are forwards, B, G and H are point guards and D, F and I are defender. **(Puzzle) NIMCET-2016**

- The team must have at least one forward, one point guard and one defender.
- If the team includes J, it must also include F.
- The team must include E or B, but not both.
- If the team includes G, it must also include F.
- The team must include exactly one among C, G and I.
- C and F cannot be members of the same team.
- D and H cannot be members of the same team.
- The team must include both A and D or neither of them.
- There is no restriction on the number of members in the team.

136. What would be the size of the largest possible team?

- (a)7 (b)6 (c)5 (d)4

137. Which of the following cannot be included in a team of size 6?

- (a)A (b)H (c)J (d)E

138. What could be the maximum size of the team that includes G?

- (a)4 (b)5
(c)6 (d)More than 6

139. Kha- kha is an obscure island which is inhabited by two types of people; the 'Yes' type and the 'No' type. Native of types 'Yes' ask only questions the right answer in which is 'Yes' while those of type 'No' ask only questions the right answer to which is 'No'. For example the 'Yes' type will ask questions like "Is 2 plus 2 equal to 4?" While the "No" type will ask question like "Is 2 plus 2 equal to 5?". The following question is based on your visit to the island Kha-kha. Kevin and Kumar are brothers from the island. Kumar asks you, Is atleast one of us is of type 'No'? You can conclude that

- (a)Kevin is 'No', Kumar is 'Yes'. **(Puzzle) NIMCET-2016**
(b)Both are 'Yes'.
(c)Kevin is 'Yes', Kumar is 'No'.
(d)Both are 'No'.

140. P, Q, R and S are four logical statements such that if P is true, then Q is true; if Q is true, then R is true; and if S is true, then at least one of Q and R is false. The it follows that

- (a) if S is false, then both Q and R true

(Puzzle) NIMCET-2016

- (b) if at least one of Q and R is true, then S is false
(c) if P is true, then S is false
(d)if Q is true, then S is false

141. A, B, C, D, E, F and G are members of a members of a family consisting of four adults, three children, three males and four female. Out of the children F and G are girls. A and D are brothers and A is doctor, E is an



engineer married to one of the brothers and has two children . B is married to D and G is their child. Who is C?

(Puzzle) NIMCET-2016

- (a) E's daughter (b) A's son
(c) G's brother (d) F's father

Directions (Q. Nos. 142 to 144) : (Puzzle) NIMCET-2016

- There are six houses P, Q, R, S, T and U, three on either side of a road.
- The houses are of different colours - red, blue, green, orange, yellow and white.
- All the houses are of different heights.
- T, the tallest house is exactly opposite to the red coloured house.
- The shortest house is exactly opposite to the green coloured house.
- U, the orange coloured house is located between P and S.
- R, the yellow coloured house is exactly opposite to P.
- Q, the green coloured house is exactly opposite to U.
- P, the white coloured house is taller than R, but shorter than S and Q.

142. Which is the second largest house?

- (a) Q (b) R
(c) S (d) Cannot be determined

143. Which of the second shortest houses?

- (a) P (b) R
(c) S (d) Cannot be determined

144. What is the colour of the tallest house?

- (a) Red (b) Blue
(c) Green (d) Yellow

145. Five children were administered psychological tests to know their intellectual levels. In the report psychologists pointed that child A is less intelligent than child B. The child C is less intelligent than child D. The child B is less intelligent than child C and child A is less intelligent than child E. Which child is most intelligent?

(Puzzle) NIMCET-2017

- (a) D only (b) E only
(c) D or E (d) Neither D nor E

146. A, B, C, D, E and F are 6 friends from a club. There are two houses wives, one lecturer, one architect, one accountant and one lawyer in the group. There are three married couples. The lawyer B is married to D, who is a houses wife. No lady is either an architect or an accountant. C, the accountant is married to F, who is lecturer. If E is not a house wife, what is the profession of E?

(Puzzle) NIMCET-2017

- (a) Lawyer (b) Architect
(c) Lecturer (d) Accountant

147. Four passengers in a train find that they from an interesting group. Two of them are lawyers and the other two are doctors. Two of them speak Bengali and the other two speak Hindi and no two of the same profession speak the same language. They also found that two of them are Christians and two are Muslims and no two of

the same religion speak the same languages. The Hindi speaking doctor is a Christian Then, which of the following statement logically follows?

(Puzzle) NIMCET-2017

- (a) The Bengali speaking lawyer is a Muslim
(b) The Christian lawyers speaks Bengali
(c) The Bengali speaking doctor is a Christian
(d) The Bengali speaking doctor is a Muslim

Directions (Q.Nos. 148 to 150) Questions are based on the following. (Puzzle) NIMCET-2017

In an amusement park seven friends-Feroz, Gautam Harish, Javed, Kumar, Laxman and Mohan are deciding who will ride the roller coaster. There is time for only one ride before the park closes.

- If Feroz rides Gautam must ride.
- If Gautam and Harish both ride, Javed cannot ride.
- If Harish and Javed both ride, Laxman cannot ride.
- If Javed rides, either Kumar or Mohan must ride.
- Kumar and Laxman cannot both ride, but one of them must ride.
- Kumar and Mohan cannot both ride.

148. Which of the following is an acceptable combination of riders if only three people ride?

- (a) Harish, Javed and Laxman
(b) Harish, Javed and Kumar
(c) Feroz, Gautam and Javed
(d) Gautam, Kumar and Laxman

149. If Javed and Mohan both ride, which of the following is true?

- (a) Gautam cannot ride (b) Harish must ride
(c) Feroz cannot ride (d) Laxman must ride

150. If Feroz and Harish both ride, what is the greatest number of people who can ride?

- (a) 5 (b) 7 (c) 4 (d) 6

151. Three thieves rob a bakery of bread, one after the other. Each thief takes half of what is present and half bread. If 3 breads remain at the end, what is the number of breads that were present initially?

(Puzzle) NIMCET-2017

- (a) 24 (b) 31 (c) 37 (d) 41

Directions (Q. Nos.152 to 156) Questions are based on the following information

There are six teachers A, B, C, D, E and F in a school. Each teacher has to teach two subjects, one compulsory and the other optional.

D's optional is History, while three other have it as compulsory subject. E and F have Physics as one of their subjects. F's compulsory subject is Mathematics, which is an optional subject of both C and E.

History and English are A's subjects but in term of compulsory and optional subjects, they are reverse of D's. Chemistry is an optional subject of one of the teachers. There is only one female teacher, who has English as her compulsory subject.





152. What is C's compulsory subject? **(Puzzle) NIMCET-2017**
 (a) Physics (b) Chemistry
 (c) English (d) History
153. Who among the following has Chemistry as a subject?
 (a) A (b) B (c) C (d) D
154. Which of the following groups of teachers has History as the compulsory subjects?
 (a) B, C and D (b) C and D
 (c) A, B and C (d) A, C and D
155. Disregarding which is compulsory or optional subject who has the same two subject combination as that of F?
 (a) B (b) E (c) D (d) A
156. How many minimum number of colours will be required to paint all the sides of a cube without the adjacent sides having the same colours? **(Puzzle) NIMCET-2017**
 (a) 3 (b) 4 (c) 5 (d) 6
157. One New York publisher has estimated that 50000 to 60000 people in the united States want an anthology that includes the complete works of William Shakespeare. And what accounts for this renewed interest in Shakespeare? As scholars point out, his psychological insights into both male and female characters are amazing even today.
 This paragraph best supports the statement that
 (a) Shakespeare 's characters are more interesting than fictional characters today
 (b) People even today are interested in Shakespeare 's work because of the character
 (c) Academic scholar are putting together an anthology of Shakespeare 's work
 (d) New Yorkers have a renewed interested in the work of Shakespeare
- (Puzzle) NIMCET-2018**
- Directions (Q. Nos. 158 to 161)** Questions are based on the following passage:
 Nine individuals Z, Y, X, W, V, U, T, S and R are the only candidates who can serve on three committees A, B and C and each candidate should serve on exactly one of the committees. Committee A should consist of exactly one member more than Committee B. It is possible that there are no members of Committee C. Among Z, Y and X none can serve on Committee A. Among W, V and U none can serve on Committee B. Among T, S and R none can serve on Committee C.
- (puzzle) NIMCET-2018**
158. In case T and Z are the individuals serving on Committee B, how many of the nine individuals should serve on Committee C?
 (a) 3 (b) 4 (c) 5 (d) 6
159. In case, T, S and X are the only individuals serving on committee B, the membership of Committee C should be
 (a) Z and Y (b) Z and W (c) Y and V (d) X and V
160. In case R is the only individual serving on Committee B, which among the following should serve on Committee A?
 (a) Wand S (b) V and T
 (c) U and S (d) T and S
161. Three persons A, B and C wear shirts of Black, Blue and Orange colours (not necessarily in the order) and pants of Green, Yellow and Orange (not necessarily in that order). No person

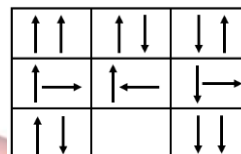
wore shirt and pant of the same colour. Further, it is given that, **(Puzzle) NIMCET-2018**

1. A did not wear shirt of Black colour.
2. B did not wear shirt of Blue colour.
3. C did not wear shirt of Orange colour.
4. A did not wear the pant of Green colour.
5. B wore pant of Orange colour.

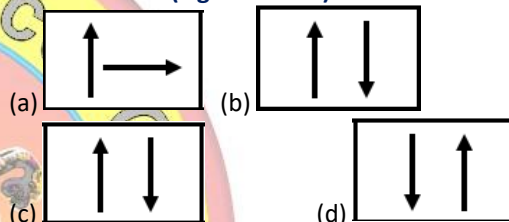
What were the colours of the pant and shirt worn by C respectively?

- (a) Orange and Black (b) Green and Blue
- (c) Yellow and Blue (d) Yellow and Black

162.



Which one of the following figures fits into blank part of the above matrix? **(Figure Puzzle) NIMCET-2018**

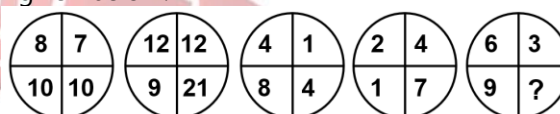


163. Forty students watched films A, B and C over a week. Each students watched either only one film or all three. Thirteen students watched film A, sixteen students watched film B and nineteen students watched film C.

How many students watched all three films? **(Puzzle) NIMCET-2018**

- (a) 0 (b) 2 (c) 4 (d) 8

164. Which number replaces the question mark in the figure given below?



- (a) 11 (b) 6 (c) 3 (d) 21

(Puzzle) NIMCET-2019

165. Each family in a locality has at most adults, and no family has fewer than 3 children. Considering all the families together, there are more adults than boys, more boys than girls, and more girls than families. Then, the minimum possible number of families in the locality is

(Puzzle) NIMCET-2019

- (a) 4 (b) 3 (c) 2 (d) 5

Directions (Q. Nos. 166 and 167) for questions 61 and 62 Answer the questions on the basis of the information given below :

A, B, C, D, E and F are a group of friends. There are two housewives, one professor, one engineer, one accountant and one lawyer in the group. There are only two married couples in the group. The lawyer is married to D, who is a housewife. No woman in the group is either an engineer or an accountant. C, the accountant, is married to F, who is a professor. A is married to a housewife. E is not a housewife.





(Puzzle) NIMCET-2019

166. What is E's profession?
(a) Accountant (b) Lawyer
(c) Professor (d) Engineer
167. How many members of the group are males?
(a) 2 (b) 3
(c) 4 (d) Cannot be determined
168. A dealer offers sells half of the eggs that he has and another half an egg to Anurag. Then, he sells half of the balance eggs and another half an egg to Deepak. Then, he sells half of the balance eggs and another half an egg to Sivani. In the end he is left with just 7 eggs and he claims that he never broke an egg. How many eggs did he start with?
(Puzzle) NIMCET-2019
(a) 65 (b) 63 (c) 67 (d) 69
169. There are eight poet, namely, P, Q, R, S, T, U, V and W and H in respect of whom questions are being asked in the examination. The first four are ancient poets and the last four are modern poets. The question on ancient and modern poets is being asked in alternate years. Those who like W also like V, those who like S like R also. The examiner who sets question is not likely to ask question on S because he has written an article on him. But he likes S. Last year a question was asked on U. Considering these facts, on whom the question is most likely to be asked this year?
(Puzzle) NIMCET-2019
(a) Q (b) R (c) S (d) V
170. A team must be selected from ten probable's-A, B, C, D, E, F, G, H, I and J. Of these, A, C, E and J are forwards, B, G and H are point guards and D, F and I are defenders. The team must have at least one forward, one point guard and one defender. If the team includes J, it must also include F. The team must include E or B, but not both. If the team includes G, it must also include F. The team must include exactly one among C, G and I. C and F cannot be members of the same team. D and H cannot be members of the same team. The team must include both A and D or neither of them. There is no restriction on the number of members in the team. What could be the maximum size of the team that includes G? **(Puzzle) NIMCET-2019**
(a) 4 (b) 5
(c) 6 (d) More than 6
171. In a family of six persons, there are people from three generations. Each person has separate profession and also they like different colours. There are two couples in the family. Rohan is a CA and his wife neither is a doctor nor likes green colour. Engineer likes red colour and his wife is a teacher. Mohini is mother in law of Savita and she likes orange colour. Deepak is grandfather of Titu and Titu, who is a principal, likes black colour. Nerru is granddaughter of Mohini and she likes blue colour. Nerru's mother likes white colour. Savita is a **(Puzzle) NIMCET-2019**
(a) Doctor (b) Teacher

- (c) Housewife (d) None of these

172. Ten points are marked on a straight line and eleven points are marked on another straight line. How many triangles can be constructed with vertices from among the above points? **(Puzzle) NIMCET-2019**
(a) 495 (b) 550 (c) 1045 (d) 2475
173. Four friends A, B, C and D need to cross a bridge in the night. A maximum of 2 people can cross at a time. They have only one lamp. A takes one min to cross the bridge. B Take 2 min, C takes 8 min and D takes 11 min to cross the bridge respectively. A pair must walk together at the speed of the person who walks slowly. What is the minimum time required to cross the bridge by all the four people? **(Puzzle) NIMCET-2020**
(a) 23 min (b) 20 min (c) 18 min (d) 16 min
174. Four students A, B, C and D distributed 30 marbles among themselves. No two students got equal number of marbles. No student got more than 10 marbles. No student got less than 5 marbles. A and C got odd number of marbles. B and D got even number of marbles. A got more marbles than B, C got more marbles than D, B got more marbles than D.
(Puzzle) NIMCET-2020
What is the number of marbles with A?
(a) 6 (b) 7 (c) 8 (d) 9
175. Four students A, B, C and D distributed 30 marbles among themselves. No two students got equal number of marbles. No student got more than 10 marbles. No student got less than 5 marbles. A and C got odd number of marbles. B and D got even number of marbles. A got more marbles than B, C got more marbles than D, B got more marbles than D.
(Puzzle) NIMCET-2020
Mean of number of marbles with B, C, D is
(a) 6 (b) 7 (c) 8 (d) None
176. Nine individuals-Z, Y, X, W, V, U, T, S and R- are the only candidates, who can serve on three committees-k₁, k₂ and k₃, and each candidate should serve on exactly one of the committees, Committee k₁ should consist of exactly one number more than committee-k₂. It is possible that there are no members in committee k₃. Among Z, Y and X none can serve on committee k₁. Among W, V and U none can serve on committee k₂. Among T, S and R none can serve on committee k₃. In case committee K₂ is served by T and Z only, how many of the nine individuals should serve on committee k₃?
(Puzzle) NIMCET-2020
(a) 3 (b) 4 (c) 5 (d) 6
177. Nine individuals - Z, Y, X, W, V, U, T, S and R- are the only candidates, who can serve on three committees- K₁, K₂ and K₃, and each candidate should serve on exactly one of the committees. Committee K₁ should consist of exactly one





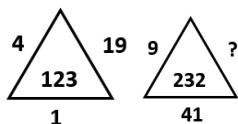
member more than committee K_2 . It is possible that there are no members in committee K_3 . Among Z, Y and X none can serve on committee K_1 , Among W, V and U none can serve on committee K_2 . **(Puzzle) NIMCET-2020**

Among T, S and R none can serve on committee K_3 . Of the nine individuals, the largest number that can serve together on committee K_3 is

- (a)8 (b)7 (c)6 (d)5

178. Find out the missing number

(Figure Puzzle) NIMCET-2020



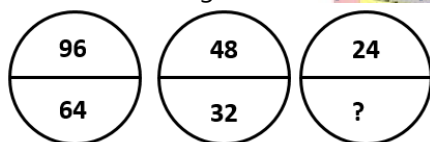
- (a)2 (b)6 (c)4 (d)16

179. Which number should come in place of the question mark(?) in the following chart: **(Figure Puzzle) NIMCET-2020**

1	7	9
2	14	?
3	105	177

- (a)16 (b)26 (c)20 (d)12

180. Find the missing number **(Puzzle) NIMCET-2020**



- (a)21 (b)16 (c)10 (d)8

181. How many minimum numbers of colours will be required to paint all the sides of a cube without the adjacent sides having the same colours? **(Puzzle) NIMCET-2020**

- (a)3 (b)4 (c)5 (d)6

182. Read the following information carefully and answer the questions given below **(Puzzle) NIMCET-2020**

- Five friends Amar, Kapil, Sarvesh, Rohan and Nagesh put on five shirts of different colour, i.e., Red, Yellow, Blue, White, and Green, while they were going to attend to party. These colours are not in the order.
- They have different hobbies as reading, playing, outing singing and writing.
- Kapil, who like singing, does not wear Yellow shirt. Sarvesh wear Red shirt and he does not like reading or writing. Nagesh likes playing and he does not wear Blue or Yellow shirt. Amar likes writing and Rohan does not wear Yellow or Green shirt.

What is the colour of Kapil's shirt?

- (a)White (b)Green
(c)Blue (d)Insufficient data to answer

183. Read the following information carefully and answer the questions given below **(Puzzle) NIMCET-2020**

- Five friends Amar, Kapil, Sarvesh, Rohan and Nagesh put on five shirts of different colour, i.e., Red, Yellow,

Blue, White, and Green, while they were going to attend a party. These colours are not in a order.

- They have different hobbies as reading, playing, outing, singing and writing.
- Kapil, who likes singing, does not wear Yellow shirt, Sarvesh wear Red shirt and he does not like reading or writing. Nagesh likes playing and he does not wear Blue or Yellow shirt. Amar likes writing and Rohan does not wear Yellow or green shirt.

Who likes writing?

- (a) Rohan (b)Amar
(c) Kapil (d) Insufficient data to answer

184. Six books are labelled A, B, C, D, E and F are placed side-by-side. Books B, C, E and F have green covers while others have yellow covers. Book A, B and D are new while the rest are old volumes. Book A, B and C are law reports while the rest are medical extracts. Which two books are old medical extracts and have green covers? **(Puzzle) NIMCET-2022**

- (a)B and C (b)E and F (c)C and F (d) C and E

Directions (Q. Nos. 185-189) : Read the following information carefully and answer the questions. **(Puzzle) NIMCET-2022**

Fives dramas A, B, C, D and E have to be staged in 6th where 1h needs to be given per drama.

- A break of 1h has to be taken in 3rd or 4th hour.
- Drama show cannot be started with A and cannot end C.
- D has to follows B immediately with no break in between.
- A cannot be done immediately after D.
- A has to precede E immediately with no break in between.

185. Which hour is a break hour?

- (a)5th (b)2nd (c)3rd (d)4th

186. Which is the drama to be staged first?

- (a)A (b)D (c)B (d)None

187. Which is the drama staged immediately after the break?

- (a)A (b)D (c)B (d)None

188. Which drama is staged immediately after D?

- (a)None (b)E (c)B (d)C

189. Which drama is staged immediately after E?

- (a)C (b)A (c)E (d)None

190. A junior school is offering five after- school activities – Karate, Handwork, Music, Dance and gymnastics. Each of the five students – Leena, Megan, Neha, Omar, and Pixie, has subscribed to at least one activity. As per the school rules, anyone who subscribes to gymnastics must also subscribe to Dance. Karate and handwork must always be subscribe together. Music and Dance cannot be subscribed together. **(Puzzle) NIMCET-2023**

The following information is available about students' subscriptions.

Megan subscribed to four activities.

Leena subscribed to gymnastics but not karate.

Pixie subscribed to only one activity and is the only one to subscribed to that activity.

Omar subscribed to three activities.

Neha subscribed to only one activity.





How many activities are subscribed by exactly two people each?

- (a)1 (b)4 (c)2 (d)3

191. In the reality show, two judges Independently provided marks based on the performance of the Participants. If the marks provided by the second judge are given by $y = 1 + x$, where x is the marks provided by the first judge. Then, for a participant.

(Puzzle) NIMCET-2023

- (a) Ranks given by both the judges differ by 2
(b) Rank given by the second judge is more than that of the first judge
(c) Ranks given by both the judge are same
(d) Rank given by the first judge is more than that of the second judge

192. Rajesh will not go to the concert if Rakesh goes. Rakesh will go to the concert if his dog barks three times. Based only on the information above, which of the following must be true? **(Puzzle) NIMCET-2024**

- (a) Rakesh will not go to the concert unless Rajesh goes.
(b) If Rakesh's dog does not bark three times, then Rakesh will not go to the concert
(c) If Rakesh's dog barks three times, then Rajesh will not go to the concert
(d) If Rajesh doesn't go to the concert, then Rakesh will go

193. You are on an island with two tribes. One tribe always tells the truth, and the other tribe always lies. You meet three individuals from the island A, B and C. Each individual belongs to one of the tribes. You ask each of them the same question "Is B a truth-teller?"

(Puzzle) NIMCET-2024

A says, "Yes B is a truth-teller."

B says, "No, I am not a truth-teller"

C. Says, "B is a liar"

Given that each individual is either a truth-teller or a liar, who is telling the truth?

- (a) A only (b) Both B and C
(c) C only (d) B only

194. In a tournament, many teams participated. All teams in the tournament have 5 to 15 players. If a team has more than 10 players, then they have reversible t-shirts?

- (a) Teams with 8 players do not have reversible t-shirts
(b) Teams that have 12 players do not have reversible t-shirts.
(c) Only people on teams can have reversible t-shirts.
(d) Teams that have 13 players have reversible t-shirts.

(Puzzle) NIMCET-2024

195. Four friends, Aditi, Bharat, Chandan, and Deepika, went to a restaurant for dinner. Each of them ordered a different dish from the menu: pizza, pasta, burger, and salad. Additionally, each friend ordered a different drink: cola, lemonade, orange juice, and water. Based on the

following clues, determine the combination of friend, dish, and drink: **(Puzzle) NIMCET-2024**

- Aditi didn't order pizza or cola.
- Bharat ordered salad but not lemonade.
- Chandan ordered pasta.
- Deepika didn't order burger or orange juice.
- Aditi ordered orange juice.

Who ordered the burger, and what drink did they order?

- (a) Deepika, cola (b) Bharat, water
(c) Aditi, orange juice (d) Chandan, lemonade

Comprehension: (196-198)

Directions: A, B, C, D, E, F, and G are travelling in three different vehicles. There are at least two passengers in each vehicle-Swift, Creta, Nexon, and only one of them is a male. There are two engineers, two doctors and three teachers among them.

- i. C is a lady doctor and she does not travel with the pair of sisters A and F.
- ii. B, a male engineer, travels with only G, a teacher in a Swift.
- iii. D is a male doctor.
- iv. Two persons belonging to the same profession do not travel in the same vehicle.
- v. A is not an engineer and travels in a Creta.
- vi. The pair of sisters A and F travel in the same vehicle.

(Puzzle) NIMCET-2024

196. What is F's profession?

- (a) Doctor (b) Data inadequate
(c) Teacher (d) Engineer

197. In which vehicle does C travel?

- (a) Creta (b) Swift
(c) Nexon (d) Data inadequate

198. Which of the following represents the three teachers?

- (a) Data inadequate (b) GEA
(c) GEF (d) GBF

Direction : A, B, C, D and E are five different integer. When written in the ascending order of values, the difference between any two adjacent integers is 8. D is the greatest and A the least. B is greater than E but less than C. The sum of the integer is equal to E.

199. The value of A is **(Puzzle) NIMCET-2024**

- (a) -18 (b) -15
(c) none of these (d) -17

200. the sum of A and B is

- (a) -30 (b) -15
(c) None of these (d) -20

201. The greatest number has the value

- (a) 14 (b) 12
(c) 15 (d) 17

202. The sum of the integers is

- (a) None of these (b) -10
(c) -8 (d) -6



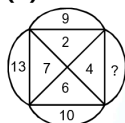


ANSWER

1	b	2	d	3	b	4	c	5	a
6	c	7	b	8	a	9	b	10	c
11	d	12	a	13	d	14	d	15	d
16	b	17	a	18	b	19	d	20	a
21	c	22	d	23	c	24	a	25	a
26	a	27	a	28	c	29	a	30	a
31	b	32	a	33	a	34	c	35	b
36	d	37	b	38	d	39	d	40	c
41	a	42	b	43	c	44	b	45	a
46	b	47	a	48	c	49	d	50	b
51	b	52	c	53	c	54	a	55	a
56	b	57	d	58	c	59	b	60	c
61	a	62	d	63	d	64	b	65	a
66	a	67	c	68	d	69	c	70	d
71	a	72	b	73	b	74	a	75	a
76	c	77	b	78	b	79	d	80	d
81	b	82	d	83	c	84	c	85	a
86	b	87	c	88	b	89	b	90	c
91	c	92	d	93	b	94	c	95	a
96	d	97	b	98	a	99	c	100	d
101	d	102	d	103	a	104	a	105	b
106	c	107	b	108	c	109	d	110	c
111	d	112	b	113	d	114	b	115	c
116	d	117	a	118	d	119	c	120	a
121	a	122	d	123	b	124	b	125	b
126	d	127	a	128	a	129	c	130	a
131	b	132	a	133	c	134	c	135	d
136	a	137	b	138	c	139	a	140	c
141	b	142	d	143	b	144	b	145	c
146	b	147	b	148	b	149	d	150	a
151	b	152	d	153	b	154	c	155	b
156	a	157	b	158	b	159	a	160	d
161	b	162	b	163	d	164	c	165	b
166	d	167	b	168	b	169	b	170	c
171	d	172	c	173	c	174	d	175	b
176	b	177	c	178	c	179	d	180	b
181	a	182	d	183	b	184	b	185	c
186	c	187	d	188	a	189	d	190	d
191	b	192	c	193	c	194	d	195	c
196	d	197	c	198	d	199	a	200	d
201	a	202	b						

SOLUTION

1. (b)



$$? = 2 + 4 = 6$$

2. (d)

Step 1.

- Green and red are opposite.
- Blue is between green and red, meaning it's adjacent to both.

Step 2.

- Yellow is adjacent to orange.

- Since green and red are opposite, and blue is adjacent to both, yellow and orange cannot be in these positions.
- Yellow and orange must occupy two of the remaining side faces.

Step 3.

- White is the only remaining colour.
- It must occupy the last remaining side face.

Step 4.

- Yellow is adjacent to orange (given).
- Yellow is also adjacent to blue and white, as they are on the side faces.
- Yellow cannot be adjacent to red or green, as they are opposite.

The colors adjacent to yellow are green, blue, white, and orange.

(3-6)

Jan	
Feb	Jayant, Chocolates
Mar	Namita, Pastries
April	Tanmay, Bengali sweets
May	

Asha does not like ice-cream

Namita does not like pry ice cream

Tanmay Bengali sweet

- (b) Dry fruits
- (c) None of these
- (a) ice cream
- (c) January or May

Qs. 7-11:

⊖	⊕	⊕	⊖	⊕	⊕	⊕	⊖
			H/C	B	D	F	C/H
-	Physics	Chemistry	Economics	Commerce	Law	Zoology	physics
Ph.d.	Ph.d.	-	-	Ph.d.	-	Ph.d.	-

7. (b) Either H or C

8. (a) G

9. (b) Two lady members are Ph.D.

10. (c) Physics - Lady - Ph.d.

11. (d) Cannot be determined

12. (a)

Step 1 Fill; 9 L container.

Step 2 Pour 4 L water from 9 L container to 4 L container.

Step 3 Empty 4 L container.

Step 4 Again pour 4 L water from 9 L container to 4 L container.

Step 5 Again empty 4 L container.

Step 6 Now pour remaining 1 L water from 9 L container to 4 L container.

Step 7 Fill 9 L container

Step 8 Pour water from 9 L container to 4 L container, so that it is completely full.

Now it is completely full.

Now, 9 L container contains 6 L water.

13. (d)





The letters are repeat ONE, TWO, THREE, FOUR and FIVE times respectively.
So, the required letter is N.

14. (d) None of the given answer is correct from the available data set.

$$G = L + 10$$

$$\frac{2A}{3} = \frac{3}{4} \left(B + \frac{A}{B} \right) = \frac{4}{5} \left(G + \frac{1}{4} \left(B + \frac{A}{3} \right) \right)$$

$$= L + \frac{1}{5} \left(G + \frac{1}{4} \left(B + \frac{A}{3} \right) \right)$$

$$\Rightarrow \frac{2A}{3} = \frac{3B}{4} + \frac{A}{4} = \frac{4G}{5} + \frac{B}{5} + \frac{A}{15}$$

$$= L + \frac{G}{5} + \frac{B}{20} + \frac{A}{60}$$

$$\Rightarrow A = 180; B = 100; G = 110; L = 80$$

Which become inconsistent.

15. (d)

If Bhalu alternated between truth and lie, so will Kachaalu, which will be a contradiction, as only one person alternated between truth and lie.

16. (b) PQSVW is not possible, as S and V cannot be together.

17. (a)

If R is in new office, then as book keeper there is only one option Q. Due to R and U in new office T and S cannot be considered. So, V and W will be sent as new secretaries. So, there is only one option QRUVW.

18. (b)

If R goes, T won't go and if S went further U and V won't go, so we will not get three secretaries.

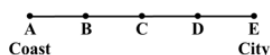
19. (d)

If S goes to new office, then team will be S, T, W, P, Q only. So, R cannot go and W must go.

20. (a)

$$\begin{array}{r} \text{S T I L L} \\ + \text{W I T H I N} \\ \hline \text{L I M I T S} \end{array} \Rightarrow \begin{array}{r} 9 \ 8 \ 5 \ 3 \ 3 \\ 2 \ 5 \ 8 \ 0 \ 5 \ 6 \\ \hline 3 \ 5 \ 6 \ 5 \ 8 \ 9 \end{array}$$

21. (c)



The person himself reach the city E from coast A, by travelling through B, C and D each at 30 km each in 4 days.

Then, when returning he will spend his rations at D, where he will be joined by second person with ration of 2 days. At C, third person will be there with ration of 3 persons for one day. Again at B, fourth person will be there, with ration of 4 persons for one day, and hence finally all 4 will reach the coast A.

22. (d)

If y was the total number of eggs and x is the daily sale. Then,

$$\text{Left over egg on day 1} = y - x$$

$$\text{Left over egg on day 2} = 2(y - x) - x = 2y - 3x$$

$$\text{Left over egg on day 3} = 3(2y - 3x) - x$$

$$= 6y - 10x$$

$$\text{Left over egg on day 4} = 4(6y - 10x) - x$$

$$= 24y - 41x$$

$$\text{Left over egg or say total eggs on day 5}$$

$$= 5(24y - 41x) = x$$

$$\Rightarrow 120y - 205 = x$$

$$\Rightarrow 120y = 206x$$

$$\Rightarrow 60y = 103x$$

$$\Rightarrow y = 103 \text{ and } x = 60 \text{ as minimum values.}$$

23. (c) Sujit has told the truth and Lakers will be the winner, so automatically remaining two statements become wrong.

24. (a)

There is continuation of series.

111 211 211 111 221 312 211 131 122 211 113 213

211 311 312 111 312 211 321 131 112 211 123 113

111 213 211 321 222 111 131 221 133

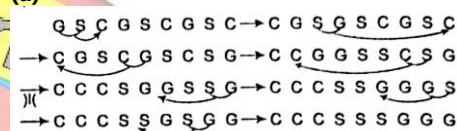
25. (a)

From the first information oldest daughter will be 9 or 12 yr old.

From the next information her age will be 9.

$$(AS \ 9 + 4 = 13)$$

26. (a)



Seven moves are shown here out of which one of them is trivial. i.e., six moves are required.

27. (a) 1st Game Mr. Birla defeated Mrs. Birla

2nd Game Mr. Birla defeated Mrs. Tata

3rd Game Mrs. Birla defeated Mr. Tata

28. (c) 15 large, 7 medium and 14 small ships

$$= 15 \times \frac{(2 \times \frac{7}{4} + 1)}{3} + 14 \text{ small ships.}$$

$$= \frac{105}{4} + \frac{21}{2} + 14 = \frac{203}{4} \text{ small ships.}$$

Now, 12 large, 14 medium and 21 small ships

$$= 12 \times \frac{7}{4} + \frac{14(2 \times \frac{7}{4} + 1)}{3} + 21 \text{ small ships}$$

$$= 21 + 21 + 21 = 63 \text{ small ships}$$

Number of trips by $\frac{203}{4}$ ships = 36

$$\text{So, number of trips by 63 ships} = \frac{\frac{203}{4} \times 36}{4} = 29 \text{ trips.}$$

29. (a) Q R T P S

According to the given information the order of reading newspaper, so Q passed newspaper to R.

Solutions (Q. Nos. 30-34)

City / Player	M	C	K	D	H
P	x	x	x	✓	x
R	x	✓	x	x	x
S	x	x	✓	x	x
U	✓	x	x	x	x
V	x	x	x	x	✓

Game / Player	Cr.	Ch.	Ca.	Ba.	T.T.
P	x	x	x	✓	x
R	x	✓	x	x	x
S	x	x	x	x	✓
U	x	x	✓	x	x
V	✓	x	x	x	✓

30. (a) P





31. (b) Hyderabad
32. (a) Badminton
33. (a) R and Chennai
34. (c)

From the given data following is the possibility.

Anil Lal Babu Sharma Bhatia Gupta

⇒ Babu has Lal and Sharma as the neighbours.

35. (b) Let x be the number of chocolates initially, Bala has.
Then, $\frac{x}{2} + 3 + \frac{1}{3}(\frac{x}{2} - 3) + 4 + \frac{1}{4}[\frac{x}{3} - 6] + 4 + 11 = x$
⇒ $\frac{3x}{4} + \frac{39}{2} = x$
⇒ $\frac{x}{4} = \frac{39}{2}$
⇒ $x = \frac{39 \times 4}{2}$
⇒ $x = 78$

36. (d) Number of direct lines
= $({}^4C_2 \times 3) \times 3 + {}^3C_2 \times {}^4C_1 \times {}^4C_1 \times 2$
= $6 \times 3 \times 3 + 3 \times 4 \times 4 \times 2$
= $54 + 96 = 150$

37. (b) Statement b weaken the argument given.
38. (d) Let x be the age of Rahul and y be the age of Preeti
⇒ $x^2 + y^3 = 7148$

$$x^3 + y^2 = 5248$$

(ii)

On solving Eqs. (i) and (ii), we get

$$x = 17 \text{ and } y = 19$$

39. (d) ? will be filled by 8.
Blocks have sum,
 $3 + 9 = 12$, $5 + ? = 13$, $7 + 7 = 14$, and $9 + 6 = 15$
40. (c) Only green rabbit can say this type of statement.
41. (a) Option (a) is the best description.
42. (b) Let x be the number of steps in staircase.
Then, $\frac{x-26}{30} = \frac{x-34}{18}$
⇒ $\frac{x-26}{5} = \frac{x-34}{3}$
⇒ $3x - 78 = 5x - 170$
⇒ $2x = 92$
⇒ $x = 46$

43. (c) In ADE ABC, only one candidate is repeated and B in there with A
44. (b) In CDE, exactly one is repeated.
45. (a) Only statement I is correct.
46. (b) Due to D there is a repetition, so rest two will be new members.
47. (a) The order of answering questions is as follows 1, 4, 7, 10, 13, 16, 19, 22, 25, 28, 31, 34, 37, 40, 43, 46, 49
1st reversal: 50, 47, 44, 41, 38, 35, 32, 29, 26, 23, 20, 17, 14, 11, 8, 5, 2
2nd reversal: 3, 6, 9, 12, 15, 18, 21, 24, 27, 30, 33, 36, 39, 42, 45, 48
48. (c) From the above last question answer is 48.
49. (d) $2017 + 3$, so 3rd question of first reversal.. 44 is the required 20th question answered by Rajita.
50. (b) In A_{20} , there will be $2 \times 20 - 1 = 39$ numbers in AP with common difference 2.
Also, the first term of A_{20} will be
 $3 + 2 \times 3 + 2 \times 5 + 2 \times 7 + \dots + 2 \times 37 + 2$
= $3 + 2(1 + 3 + 5 + 7 + \dots + 37)$

$$= 3 + 2 \times \frac{19}{2} [2 + 18 \times 2] = 3 + 2 \times 19 \times 19 = 725$$

Average of 39 numbers of A_{20}

$$= \frac{\frac{39}{2} [2 \times 725 + (39 - 1) \times 2]}{39} = 725 + 38 = 763$$

Solutions (Q. Nos. 50-54)

House	A	B	C	D	E
Roof		R			G
Chimney	W	B	W		

51. (b) A and C have white chimney.
52. (C) A and C have white chimneys and E may also have white chimney.
53. (c) A, C and E can have green roofs.
54. (a) All three combinations are possible.
55. (a) House E has a white chimney.

Solution:

56. (b)

57. (d)

58. (c)

59. (b)

Direction Qs. 8 to 9: The cost will be minimum by using type Q thrice and s one each.

60. (c) Type Q $\times 3$ + Type R $\times 1$ + Type a $\times 1 = 220$
 $140 \times 3 + 125 \times 1 + 25 \times 1 = 640$

61. (a) The number of buses required the cost is minimum is 5.
62. (d) The next best cost is obtained by using type Q two times and type R three times the cost.
= $2 \times 140 + 3 \times 125 = \text{Rs. } 655$

63. (d)

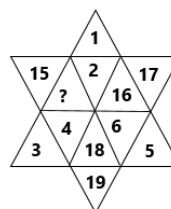
56	65	78
12	51	30
44	14	48

$$44 + 12 = 56$$

$$48 + 30 = 78$$

$$14 + x = 65 \quad x = 51 \text{ Ans.}$$

64. (b)



$$19 - 1 = 18$$

$$5 + 1 = 6$$

$$17 - 1 = 16$$

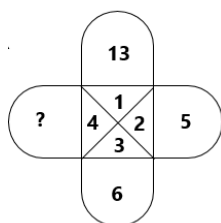
$$3 + 1 = 4$$

$$1 + 1 = 2$$

$$15 - 1 = 14$$

65. (a)





$$6 = 4 \times 1 + 2$$

$$5 = 2 \times 1 + 3 = 5$$

$$13 = 4 \times 1 + 1 = 13$$

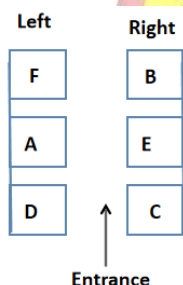
$$x = 3 \times 2 + 4 = 10$$

(66-68)

P = Horticulturist – tea
Q = Industrial – coffee
R = Physicist- tea
S = Journalist coffee
T = Advocate Tea

- 66. (a)** P is the horticulturist.
67. (c) Q is the industrialist.
68. (d) All choices the advocate = T.

Direction for Qs. 69-73



- 69. (c)** If E sits in his office and faces the corridor C's office will be to his left.
70. (d) E's office faces A's office.
71. (a) Only A is F's neighbour.
72. (b) Directing the person to B's office.
73. (b)
- | Y/w | z | w/N |
|-------|--------|-----|
| Green | yellow | Red |

- 74. (a)** C's Amount = Rs. 147.
75. (a) B's Amount = Rs. 102.
76. (c) C get the highest amount Rs. 147.
77. (b) No. 2 represent the educated, hardworking and urban people.
78. (b) No. 7 represents non-urban educated people who are neither hard working nor honest.
79. (d) Number 9 represents honest, educated and hardworking non-urban people.

Solutions [Q. Nos. 80 to 83]

From the given information, we can definitely say that B's presentation will be immediately before C's presentation and their presentation together will be either pre lunch or post lunch. D is either first or last scientist to present his paper.

- 80. (d)** If C is fifth scientist to present his paper, then B must be fourth to present his paper because B's presentation will be immediately before C's presentation.
81. (b) We know that B and C scientists presentation cannot be separated by the lunch break, in this case B could be placed in third order of presenters.
82. (d) From the given information, we can say that, D is the first scientist and F is the second scientist, then C can be present in only either fifth or sixth place. As B and C have to present in two consecutive position either pre lunch or post lunch.
83. (c) If F and E are fifth and sixth scientists, then D will be the first scientist. Band C will be the second and third presenters. Hence, A is the fourth presenter.

Solutions (Q. Nos. 84-86)

Time taken for stitching school uniforms

School	Cutting	Stitching	Button Stitching
May flower	20 min	60 min	15 min
Little flower	30 min	60 min	30 min

Number of fabric cutters = 2
Number of tailors = 5
Number of assistants = 2
Each of them do the work for 10 h.

Time duration for fabric cutters = $2 \times 10 \text{ h}$
= 20 h
= 1200 min
Time duration for tailors = $5 \times 10 \text{ h}$
= 50 h
= 3000 min
Time duration for button stitchers = $2 \times 10 \text{ h}$
= 20 h
= 1200 min

Maximum number of Little flower uniforms that can be cut in a day = $\frac{3000}{30} = 40$
Maximum number of Little flower uniforms that can be stitch in a day = $\frac{3000}{60} = 50$
Maximum number of Little flower uniforms that can be buttoned in a day = $\frac{1200}{30} = 40$

- 84. (c)** Number of Little flower uniforms that Mrs Thomas can complete in a day = 40
85. (a) If Mrs Thomas decided to complete 20 Little flower uniforms.
Then, time duration for fabric cutters
= $1200 - (20 \times 30)$
= 600 min
Time duration for tailors = $3000 - (20 \times 60) = 1800 \text{ min}$
Time duration for assistants = $1200 - (20 \times 30)$
= 600 min
Hence, number of May flower – uniforms that can be completed in a day
= minimum of $\left(\frac{600}{20}, \frac{1800}{60}, \frac{600}{15}\right)$
= minimum of (30, 30, 40) = 30
86. (b) If Mrs Thomas hires one more assistants, then time duration for button stitchers
= $1200 + (1 \times 10 \times 60) = 1800 \text{ min}$





Hence, maximum May flower uniforms that can be completed in a day

$$= \text{minimum of } \left(\frac{1200}{20}, \frac{3000}{60}, \frac{1800}{15} \right)$$

$$= \text{minimum of } (60, 50, 120) = 50$$

Solutions (Q.Nos. 87-89)

- (i) A causes B or C but not both.
- (ii) F occurs only if B occurs.
- (iii) D occurs if B or C occurs.
- (iv) E occurs only if C occurs.
- (v) C occurs only if E or F occurs
- (vi) D causes G or H or both
- (vii) H occurs if E occurs.
- (viii) G occurs if F occurs.

87. (c) From Statement (i), A causes B or C but not both. From Statement (ii), F occurs only if B occurs and from Statement (iii), D occurs if B or C occur. It means I and II may occur. From Statements (vi) and (vii), II and III are may occur. So, we conclude that I and III or II and III may occur but not both occur.

88. (b) From Statement (ii) that F occurs only if B occurs and from Statement (viii) that if G occurs if F occurs it means if B occurs G must occur.

89. (b) From Statement (v), that J occurs only if E or F occurs. From Statement (ii), F occurs only if B occurs and from Statement (iv), E occurs only if C occurs it means if J occurs either B or C must have occurs.

90. (C) $E = 6 - 9 + 8 \times \frac{3}{20}$
By given condition,
 $E = 6 + 9 \times \frac{8}{3} - 20$
 $E = 6 + 3 \times 8 - 20$
 $E = 6 + 24 - 20$
 $E = 6 + 4 = 10$

91. (c) From the statements, we clearly say that the reason behind the nearsightedness of the children is caused by the visual stress required by reading and other class work.

Solutions (Q.Nos. 92-94)

Randy	Vacuuming	Monday
Sally	Dusting	Tuesday
Terry	Sweeping	Wednesday
Uma	Mopping	Thursday
Vernon	Laundry	Friday

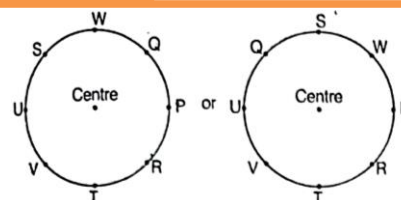
92. (d) Sweeping

93. (b) Monday

94. (c) Tuesday

Solutions (Q.Nos. 95-96)

According to the given data, we get the following figure



95. (a) R and V are not neighbours.

96. (d) The position of S is not fixed. So, data inadequate.

Solutions (Q.Nos. 97-99)

97. (b) From the above diagram, it is clear that atleast two houses have red roofs.

98. (a) If house C has a yellow roof it means house D has a red roof. So, house D never has a red chimney. So, chimney of D will be of black colour, so colour of chimney of house E will be white.

99. (c) The maximum number of green roofs are 3.

Solutions (Q. Nos. 100 to 101)

1	2	3	4	5	6
(Mr. Donald)	(Mr. Tim)	(Mr. Mike)	(Mr. Brown)	(Miss. Hardy)	(Miss. Robert)
Smoker	smoker	smoker	Non-smoker	Non-smoker	Non-smoker

100.(d) Mr. Donald's office is farthest from Mr. Brown.

101.(d) The three smoker employers of the offices should be seated in 1, 2 and 3.

102.(d) The ideal office for Mr. Mike would be 3.

Solutions [Q. Nos. 103 & 104]

According to the question.

- A = 2B ... (i)
- B = 4.5 C ... (ii)
- C = $\frac{1}{2}$ D ... (iii)
- D = $\frac{1}{2}$ E ... (iv)
- A > E > C ... (v)

Combining all the equations,

$$A > B > E > D > C$$

103.(a) A is the heaviest article in weight.

104.(a) A, B, E, D, C

Solutions [Q. Nos. 105 to 107]

- A (only ON) B (ON) ... (i)
- A (ON) + B (ON) → C (ON) ... (ii)
- A (ON) + B (ON) + C (ON) → C (OFF) ... (iii)
- Except,
- ON → OFF and OFF → ON

105.(b) If switches A and B are ON and C is OFF, their changed settings will be A ON, B ON and C ON.

106.(c) If only B is ON, the changed setting will be A ON, B OFF and C ON.

107.(b) If only B is ON in the changed setting, the original setting could have been A ON, B OFF and C ON.





108.(c) It is very clear from the given information that C does not work with A, therefore should be the partner with B as they can form pairs i.e. AD and CB.

So, according to the question,

Number of possibility for each player is

A	B	C	D	
A	x	x	x	✓
B	x	x	✓	x
C	x	x	x	✓
D	✓	x	x	x

Solutions [Q. Nos. 109 to 111]

A table is drawn on the basis of the given information.

Town	A	B	C	D	E
Team	T	R	P	Q	S

109.(d) Team Q is from town D, according to the above table.

110.(c) Team P is from the town C, according to the above table.

111.(d) Team T is from the town A, according to the above table.

112.(b) Let the weights of the five boxes, in increasing order be A, B, C, D and E.

i.e. $A < B < C < D < E$.

Each of the boxes can be paired up with another box in a total of 4 ways.

If we sum up all the given weights, we get

$$4(A + B + C + D + E) = (110 + 112 + 113 + 114 + 115 + 116 + 117 + 118 + 120 + 121)$$

$$= 1156 \text{ kg}$$

$$\Rightarrow A + B + C + D + E = 289 \text{ kg} \quad \dots(i)$$

Now, 121 kg is the sum of the weight of the boxes D and E and 110 is the sum of the weights of the boxes A and B.

i.e. $A + B = 110 \text{ kg}$

and $D + E = 121 \text{ kg}$

$$\therefore A + B + D + E = 231 \text{ kg} \quad \dots(ii)$$

Subtracting Eq. (ii) from Eq. (i), we get

$$C = 289 - 231 = 58 \text{ kg}$$

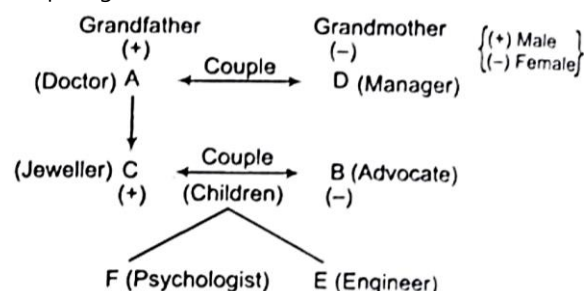
Now, 120 kg is the sum of the weights of the boxes C and E

i.e. $C + E = 120 \text{ kg}$

$$E = (120 - 58) \text{ kg} = 62 \text{ kg}$$

Solutions [Q. Nos. 113 to 117]

On the basis of the given information, we have the following 1 relationship diagram.



113.(d) Profession of A is doctor. But it is not given in any of the options.

114.(b) Profession of E is engineer.

115.(c) A is grandfather of E.

116.(d) This cannot be determined.

117.(a) AD and CB are the two couples in the family.

Solutions [Q. Nos. 118 to 120]

Based on the given information, following table is drawn.

Executive	CEO	President	Vice President	Secretary	Treasurer
Executive names	Cheryl	Bert	Enid	Alice	David
Cars	Red	Blue	Green	Yellow	Purple
Order	1	2	3	4	5

118.(d) Alice is the secretary.

119.(c) Cheryl is the CEO.

120.(a) Vice-President's car is of green colour.

Solution [Q. Nos. 121 to 124]

The given information can be represented as:

Stage	Farmer	Starting point of the stage	Ending point of the stage
First	F ₂	P ₂	P ₅
Second	F ₁	P ₅	P ₃
Third	F ₂	P ₃	P ₁
Fourth	F ₄	P ₁	P ₄
Fifth	F ₅	P ₄	P ₂

121. (a) P is the finishing point of

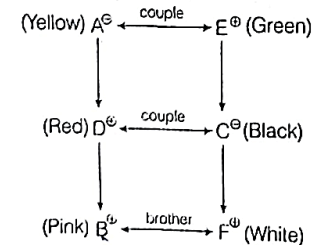
122. (d) 5th stage is ploughed by

123. (b) P and P

124. (b) P is the starting point of stage 3.

Solutions [Q. Nos. 125 to 128]

From the given information, we can draw a relation diagram,



125. (b) A likes yellow colour.

126. (d) Yellow-Hypen green is the colour combination of AE

127. (a) CD is the married couple.

128. (a) F is the brother of B.

129. (d) From the given information

Wife	Husband
X	A (Lawyer)





Y	B (Doctor)
Z	C (Engineer)

From the above table, no statement is correct.

Directions [Q. Nos. 130-133]

The given information in the question can be summarised as:

Girl	Money	Item
A	10	Lollipops
B	200	Toffees
C	60	Chocolates
D	40	Lollipops
E	120	Chocolates
F	80	Toffees
G	20	Chocolates

130 (a) From the table, it is clear that G has Rs.20.

131 (b) From the table, it is clear that C, G and E has chocolates with them.

132.(a) Combination in option (a) is correct.

133.(c) From the table, it is clear that D has Rs.40 with her.

Solution:

134. (c) Total number of fruits in the basket

$$= 19 \times 1 - 4 \times 2$$

$$= 19 - 8 = 11$$

135. (d) Total number of two's = total number of orange = 6

Total number of four = 4

$\therefore$ Total number of oranges in the basket

$$= 6 - 4 = 2$$

Solutions (Q. Nos. 136- 138)

Forwards - A, C, E and J

Point Guard - B, G and H

Defenders - D, F and I

136. (a) The largest possible team shall be of 6.

F, J, A, D, G and B/E.

137. (b) H cannot be included in 6 member team as if H is included then A and D will not be in the team.

138. (c) G will be a member of 6 member team.

139. (a) If the answer of question is 'No', then none belongs No-zone. This implies both are Yes. Which is contradiction. If Kumar is 'Yes' the answer to the questions is 'Yes' tha implies that Kevin is from No.

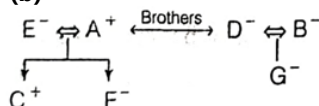
140. (c) If P is true, then Q is true.

If Q is true, then R is true.

So, if P is true, then Q and R are both true.

and if Q and R are true, then S is false.

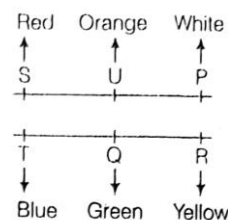
141. (b)



So, C is son of A and E.

Solutions. (Q. Nos. 142-144)

From the given information the arrangement will be as follows.



Order of height $T > S/Q > P > R > U$

142. (d) 2nd highest cannot be determined.

143. (b) 2nd shortest house is R.

144. (b) The tallest house is T which is Blue.

145. (c) According to the given information,

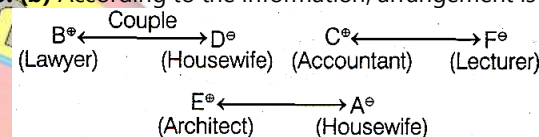
$A < B, C < D, B < C$ and $A < E$

$A < B < C < D$ and $A < E$

$\therefore$ E's position can't be determined.

Hence, most intelligent child is either D or E.

146. (b) According to the information, arrangement is as follows



E is an architect.

147. (b) According to the information, arrangement is as follow

Profession	Language	Religion
Lawyer	Bengali	Christian/Muslim
Lawyer	Hindi	Muslim
Doctor	Bengali	Muslim/Christian
Doctor	Hindi	Christian

$\therefore$ Lawyer is either Christian or Muslim, he will definitely speaks Bengali.

Hence, option (b) is the correct answer.

148. (b) From all the conditions given in the questions, Option (a) is not possible as Harish, Javed and Laxman cannot ride together given in condition (iii).

Option (c) is not possible as in condition (v), in Kumar or Laxman one person is compulsory. No one is given in option (c).

Option (d) is not possible as Kumar and Laxman cannot ride together.

Hence, option (b) is correct which fulfills all the conditions.

149. (d) If Javed rides then either Kumar or Mohan will definitely ride by condition (iv).

But, Javed and Mohan is already given. Thus, Kumar will not ride.

But in condition (v), one person from Kumar or Laxman will definitely ride. Hence, Laxman must ride.

150. (a) If Feroz and Harish both ride, then Gautam must ride. By condition (ii), Javed will not ride.

By condition (vi), either Kumar or Mohan will ride and by condition (v), either Kumar or Laxman is compulsory.

If Mohan rides, then Laxman will ride.

And, if Kumar rides then Laxman cannot ride.

Hence, Feroz, Harish, Gautam, Mohan and Laxman can ride.





151. (b) Let number of breads available be 'x'.

According to question,

Number of breads robbed by 1st thief,

$$= \frac{x}{2} + \frac{1}{2} = \frac{x+1}{2}$$

Now, number of breads left after 1st robbery,

$$= x - \frac{x+1}{2} = \frac{x-1}{2}$$

Again, number of breads robbed by 2nd thief,

$$= \frac{1}{2} \left(\frac{x-1}{2} \right) + \frac{1}{2}$$

$$= \frac{x-1}{4} + \frac{1}{2} = \frac{x+1}{4}$$

So, number of bread left after 2nd robbery,

$$= \left(\frac{x-1}{2} \right) - \left(\frac{x+1}{4} \right)$$

$$= \frac{2x-2-x-1}{4} = \frac{x-3}{4}$$

Now, number of breads robbed by 3rd thief,

$$= \frac{1}{2} \left(\frac{x-3}{4} \right) + \frac{1}{2}$$

$$= \frac{x-3}{8} + \frac{1}{2} = \frac{x+1}{8}$$

Hence, number of breads left after 3rd robbery,

$$= \left(\frac{x-3}{4} \right) - \left(\frac{x+1}{8} \right) = \frac{x-7}{8}$$

Now,

$$\frac{x-7}{8} = 3$$

$$x = 31$$

$$\Rightarrow x = 31$$

Therefore, total number of breads present initially is 31.

Sol. (Q. Nos. 152-156) According to the information, arrangement is as follows :

	Compulsory Subject	Optional Subject
A	History	English
B	History	Chemistry
C	History	Mathematics
D	English	History
E	Physics	Mathematics
F	Mathematics	Physics

152. (d) History is D's compulsory subject.

153. (b) B has Chemistry as a subject.

154. (c) Group 'A, B and C' has history as the compulsory subject.

155. (b) E has the same two subject combination as that of F.

156. (a) Three different colours are required to point all the sides of a cube so that no two adjacent sides having the same colours. Opposite faces have same colour.

157. (b) Clearly, the paragraph best supports the statement (b).

158. (b) Two members (T and Z) are serving committee B.
∴ Three members would serve committee A. and hence, 4 members would serve on committee C.

159. (a) T, S and X serve committee B.

∴ 4 members would serve committee A.

x. Now, Z, Y and X cannot serve committee A.

Hence, Z and Y serve on committee C.

160. (d) R is the only individual serving committee B.

∴ 2 members would serve committee A and the rest would serve committee C.

Now, T, S and R cannot serve on committee C.

Hence, T and S serve on committee A.

161. (b) According to the given information,

Person	Shirt's Colour	Pant's Colour
A	Orange	Yellow
B	Black	Orange
C	Blue	Green

Clearly, C wore pant and shirt of Green and Blue colour, respectively.

162. (b) In each column the element on the left hand side remains the same while the element on the right rotates 90° clock wise in each step.

163. (d) Students who watched film A = 13

Students who watched film B = 16

Students who watched film C = 19

But total number of students = 40

Hence, students who watched all the films

$$= (13 + 16 + 19) - 40 = 8$$

164. (c) With the help of lower two circles as a source, the values in corresponding segments of the upper left circle equal the sums of the numbers in the lower two circles. The values in the upper central circle equal the products of the values in the lower two circles and the upper right circle equals the difference between values in the lower two circles.

$$\therefore ? = 3$$

165. (b) Let n_f be number of families, n_g be number of girls, n_b be number of boys, n_a be number of adults. Number of adults more than number of boys, number of boys more than number of girls and number of girls more than number of families.

Hence, we have

$$n_a > n_b > n_g > n_f \quad \dots (i)$$

We check the above relation (1) with different values for number of families starting with $n_f = 1$

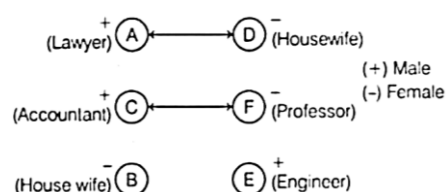
Let $n_f = 1$, then $n_b + n_g = 3$, if $n_g = 2$ then $n_b = 1$ But number of boys is greater than number of girls due to this contradiction, $n_f \neq 1$

Let $n_f = 2$, then $n_b + n_g = 6$, if $n_g = 3$, then $n_b = 3$ But number of boys is greater than number of girls due to this contradiction, $n_f \neq 2$

Let $n_f = 3$, then $n_b + n_g = 9$, if $n_g = 4$, then $n_b = 5$, number of adults $n_a = 6$ Hence, the relation $n_a > n_b > n_g > n_f$ is satisfied for the above numbers, $6 > 5 > 4 > 3$

Hence, minimum number of families is 3.

Sol. (Q. Nos. 166-168)



166. (d) Engineer

167. (b) 3

168. (b) Let he start with x eggs.

Then the amount of eggs grocer have after selling to Anurag

$$= x - \left[\frac{x}{2} + \frac{1}{2} \right] = \frac{x}{2} - \frac{1}{2}$$

Similarly after selling to Deepak 2 4 2



$$= \frac{x-1}{2} - \left[\frac{x-1}{4} + \frac{1}{2} \right]$$

$$= \frac{x-3}{4}$$

And after selling to Shivani

$$= \frac{x-3}{4} = \left[\frac{x-3}{8} + \frac{1}{2} \right] = \frac{x-7}{8}$$

Since, grocer had left with 7 eggs.

Therefore, $\frac{x-7}{8} = 7$

$\Rightarrow x = 63$

Hence, grocer had initially 63 eggs.

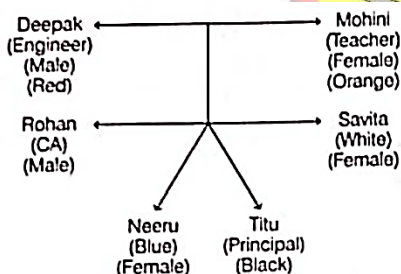
- 169. (b)** Given. Ancient poets P, Q, R, S
Modern poets T, U, V, W

Last year a question was asked on U i.e. on Modern poet. So, this year the question will be asked on ancient poet because the question on ancient and modern poets is being asked in alternate year.

It is also given, the question on S will not be asked but the examiner likes S, so he also likes R. Hence, the question is most likely to be asked this year on R.

- 170. (c)** The maximum members of the team which includes G can be 6 i.e. A, D, G, F, E, J. or A, D, G, F, B, J

- 171. (d)** We have,



Since, no information is given regarding profession of Savita. So, option (d) is correct.

- 172. (c)** Triangle is formed by picking 3 point 2 from 1st line and 1 from 2nd or 1 point from 1st line or 2 from 2nd.

i.e. ${}^{10}C_2 \times {}^{11}C_1 + {}^{10}C_1 \times {}^{11}C_2$

$$= \frac{10!}{2!(10-2)!} \times \frac{11!}{1!(11-1)!} + \frac{10!}{1!(10-1)!} \times \frac{11!}{2!(11-2)!}$$

$$= \frac{10!}{2!8!} \times \frac{11!}{1!10!} \times \frac{10!}{1!9!} \times \frac{11!}{2!9!}$$

$$= \frac{10 \times 9}{2 \times 1} \times \frac{11}{1} + \frac{10}{1} \times \frac{11 \times 10}{2 \times 1}$$

$$= 45 \times 11 + 10 \times 55$$

$$= 495 + 550$$

$$= 1045$$

- 173. (c)**

First A and B will go in 2 min, then A returns in 1 min. Then C and D will go in 11 min. B returns in 2 min. A & B will go in 2 min. The total time taken would be $(2 + 1 + 11 + 2 + 2) = 18$ min.

- 174. (d)** Given 4 students A, B, C, D

$\Rightarrow$ They distributed 30 marbles among themselves.

- (a) No two students got equal number of marbles.
(b) No student got more than 10 marbles.
(c) No student got less than 5 marbles.
(d) A and C got odd number of marbles. It can be 7 or 9.
(e) B and D got even number of marbles. It can be 6, 8 or 10.
(f) A got more marbles than B, C got more marbles than D
 $A > B > D; C > D$

$A(7,9) > B(6,8) > D(6,8), C(7,9) > D(6,8)$

So, D got the least number of marbles.

$\therefore$ D got 6 marbles.

So, B got 8 marbles as B got more marbles than D.

$\Rightarrow$ A got 9 marbles as A got more marbles than B.

$\Rightarrow$ C got 7 marbles as C got more marbles than D.

Students	Definite Number of marbles
A	9
B	8
C	7
D	6

$\therefore$ The number of marbles with A is 9.

$\therefore$ The correct answer is option (d).

- 175. (b)** As per the question, the different number of marbles.

Students	Definite Number of marbles
A	9
B	8
C	7
D	6

Mean of number of marbles with B, C, D is

$$\frac{8+7+6}{3} = \frac{21}{3} = 7$$

$\therefore$ The correct answer is option (b).

- 176. (b)** Given, Nine individuals : Z, Y, X, W, V, U, T, S and R.

Three committees : K₁, K₂ and K₃

- Each candidate should serve on exactly one of the committees.
- Among Z, Y and X none can serve on committee K₁.
- Among T, S and R none can serve on committee K₃.
- Among W, V and U none can serve on committee K₂.

In case committee K₂ is served by T and Z.

Committees	Individual who can serve	Individuals who can't serve
K ₁		Z, Y, X
K ₂	T, Z	W, V, U
K ₃		T, S, R

K₁ should consist of exactly one member more than committee K₂.

So, K₁ is served by three members as K₂ is served by two members only.

So, remaining individuals will serve committee K₃.

$$\text{Remaining persons} = 9 - (2 + 3)$$

$$= 9 - 5 = 4$$

The correct answer is option (b).

- 177. (c)** As per the given question,

Committees	Individual who can serve	Individuals who can't serve
K ₁		Z, Y, X
K ₂		W, V, U
K ₃		T, S, R

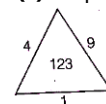
It is possible that there are no members in committee K₃.

Of the nine individuals, the largest number that can serve together on committee K₂ = 9 - Number of members who can't serve K₃.

$$= 9 - 3 = 6$$

$\therefore$ The correct answer is option (c).

- 178. (c)** As per the given question.





⇒ It follows the sequence.

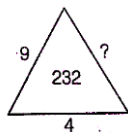
$$1 \rightarrow 1^2$$

$$4 \rightarrow 2^2$$

$$9 \rightarrow 3^2$$

So, in triangle it is written as 123.

In the same way.



⇒ It also follows the sequence

$$4 \rightarrow 2^2$$

$$9 \rightarrow 3^2$$

$$4 \rightarrow 2^2$$

So the missed term is 4.

∴ The correct answer is option (c).

179. (d) Given question

1	7	9
2	14	?
3	105	117

The pattern follows in the series is:

$$\Rightarrow (2 \times 1 + 1) = (2 + 1) = 3$$

$$\Rightarrow (7 \times 14 + 7) = (98 + 7) = 105$$

⇒ In the same way

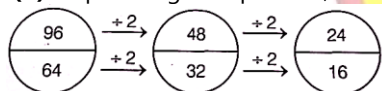
$$(9 \times ? + 9) = 117$$

$$9x + 9 = 117$$

$$9x = 108 \Rightarrow x = 12$$

∴ The correct answer is option (d).

180. (b) As per the given question,



∴ The correct answer is option (d).

181. (a) Cube has 6 sides.

As no two adjacent sides have the same colour, any pair of two opposites sides can have the same colour.

Hence, minimum number of colour required = $\frac{6}{2} = 3$.

∴ The correct answer in option (a).

182. (d) Five friends Amar, Kapil, Sarvesh, Rohan and Nagesh.

Different colours Red, Yellow, Blue, White, Green

(White they were going in attend a party. These colours are not in order)

Different Hobbies: Reading, Playing, Outing, Singing and Writing.

1. Kapil, who likes singing, does not wear Yellow shirt.
2. Sarvesh wears Red Shirt and he does not like reading or writing.
3. Nagesh likes playing and he does not wear Blue or Yellow shirt.
4. Amar likes Writing and Rohan does not wear Yellow or green shirt.

Friend	Colour	Hobbies
Amar		Writing
Kapil		Singing
Sarvesh	Red	
Rohan		
Nagesh		Playing

Amar wears Yellow shirt and Rohan likes Reading.

Sarvesh likes outing.

The colour of Kapil's shirt may be Green or Blue or White.

The colour of Rohan's shirt may be white or Blue.

The colour of Nagesh's shirt may be white or Green.

Rohan likes Reading

Friend	Colour	Hobbies
Amar	Yellow	Writing
Kapil	Green/Blue/White	Singing
Sarvesh	Red	Outing
Rohan	Blue/White	Reading
Nagesh	Green/White	Playing

The colour of Kapil's shirt can't be answer (d) as data is insufficient.

∴ The correct answer is option (d).

183. (b) The arrangement as per data given is

Friend	Colour	Hobbies
Amar	Yellow	Writing
Kapil	Green/Blue/White	Singing
Sarvesh	Red	Outing
Rohan	Blue/White	Reading
Nagesh	Green/White	Playing

∴ Amar likes writing.

∴ The correct answer is option (b).

184. (b) According to the question,

Qualities/Bodes	Green cover	Yellow cover	New volume	Old volume	Law reports	Median extract
A	✓	✓	✓	×	✓	×
B	×	×	✓	×	✓	×
C	✓	×	×	✓	✓	×
D	×	✓	✓	×	×	✓
E	✓	×	×	✓	×	✓
F	✓	×	×	✓	×	✓

Hence, E and F are two old medical extracts and have covers.

Solutions (Q. Nos. 185-189)

According to the given information

Hours	Dramas
1 st	B
2 nd	D
3 rd	Break
4 th	C
5 th	A
6 th	E

185. (c) 3rd hour is a break hour.

186. (c) Drama B is to be staged first.

187. (d) Drama C is to be staged immediately after the break.

188. (a) Break hour happens immediately after D.

189. (d) Drama E is to be staged last.

190. (d) According to the question,

	Karate	Handwork	Music	Dance	Gymnastic
Megan	✓	✓	×	✓	✓
Leena	×	×	×	✓	✓
Pixie	×	×	✓	×	×
Omar	✓	✓	×	✓	×
Neha	×	×	×	✓	×





It can clearly seen that there are three activities (karate, Handwork and Gymnastic) subscribed by exactly two people each.

191. (b) According to the question,

Given that, marks provided by th second judge are given by $y = 1 + 4$,

It means the marks given by y is 1 more than x .

So, ranks given by the second judge is more than that of the first judge.

192.(c)

193.(c)

A	✓	x	truth
B	x	x	
C	✓	✓	

194.(d)

195.(c)

A	B	C	D
Buger	Salad	Pasta	Pizza
Orange Juice	-	-	-

196.(d)

197. (c)

198. (d)

199.(a)

$$\begin{aligned}
 X + 32 &= D + 14 \\
 X + 34 &= C + 6 \\
 X + 16 &= B - 2 \\
 X + 8 &= E - 10 \\
 Z &\leftarrow A - 18 \\
 A + E + B + C + D &= E \\
 X + x + 16 + x + 24 + x + 32 &= 0 \\
 4x &= -72 \\
 x &= -18
 \end{aligned}$$

200. (d)

$$\begin{aligned}
 A + B \\
 x + x + 16 \\
 2x + 16 \\
 -36 + 16 \\
 = -20
 \end{aligned}$$

201. (a)

$$\begin{aligned}
 x + 32 \\
 -18 + 32 \\
 = 14
 \end{aligned}$$

202. (b)

$$\begin{aligned}
 x + 8 \\
 -18 + 8 \\
 = -10
 \end{aligned}$$

